

Ibuprofen: Drug information - UpToDate

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For additional information see "Ibuprofen: Patient drug information" and "Ibuprofen: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Serious cardiovascular thrombotic events (excluding NeoProfen):

Nonsteroidal anti-inflammatory drugs (NSAIDs) cause an increased risk of serious cardiovascular thrombotic events, including myocardial infarction, and stroke, which can be fatal. This risk may occur early in treatment and may increase with duration of use.

Ibuprofen is contraindicated during the perioperative setting of coronary artery bypass graft (CABG) surgery.

Serious gastrointestinal bleeding, ulcerations, and perforation (excluding NeoProfen):

NSAIDs cause an increased risk of serious gastrointestinal (GI) adverse events including bleeding, ulceration, and perforation of the stomach or intestines, which can be fatal. These events can occur at any time during use and without warning symptoms. Elderly patients and patients with a prior history of peptic ulcer disease and/or GI bleeding are at greater risk for serious GI events.

Brand Names: US

- Advil [OTC];
- Advil Junior Strength [OTC];
- Advil Liqui-Gels minis [OTC];
- Advil Migraine [OTC];
- Advil [OTC];
- Caladolor;
- Childrens Advil [OTC];
- Childrens Ibuprofen [OTC];
- Childrens Motrin [OTC];
- FT Ibuprofen Childrens [OTC];
- FT Ibuprofen IB Childrens [OTC];
- FT Ibuprofen Infants [OTC];
- FT Ibuprofen Minis [OTC];
- FT Ibuprofen [OTC];
- FT Pain Relief [OTC];
- GoodSense Ibuprofen Childrens [OTC];

- GoodSense Ibuprofen [OTC];
- I-Prin [OTC] [DSC];
- IBU;
- Ibuprofen Childrens [OTC];
- Infants Advil [OTC];
- Medi-First Ibuprofen [OTC];
- Motrin Childrens [OTC];
- Motrin IB [OTC];
- Motrin Infants Drops [OTC];
- NeoProfen;
- Proprinal [OTC]

Brand Names: Canada

- APO-Ibuprofen;
- TEVA-Profen [DSC]

Pharmacologic Category

- Analgesic, Nonopioid;
- Nonsteroidal Anti-inflammatory Drug (NSAID), Oral;
- Nonsteroidal Anti-inflammatory Drug (NSAID), Parenteral

Dosing: Adult

Dosage guidance:

Safety: Avoid or use with caution in patients at risk for or with existing cardiovascular disease, GI disease, kidney impairment, chronic liver disease, or a bleeding diathesis. Consider proton pump inhibitor coadministration in patients at risk for GI bleeding (eg, taking dual antiplatelet therapy or an anticoagulant, ≥ 60 years of age, high doses) (Ref).

Dosing: Use the lowest effective dose for the shortest duration of time.

Abnormal uterine bleeding, nonacute

Abnormal uterine bleeding, nonacute (alternative agent) (off-label use): **Note:** Not indicated for management of acute abnormal bleeding (ie, excessively heavy or prolonged bleeding that requires urgent evaluation). Alternative for patients who cannot or choose not to use hormonal therapies (Ref). Dosing based on limited data and expert opinion.

Oral: 600 mg 2 to 4 times daily. Begin at menses onset and continue for 2 to 3 days or until cessation of bleeding (Ref).

Anti-inflammatory

Anti-inflammatory (eg, for arthritis associated with rheumatic disease):

Oral: 400 to 800 mg every 6 to 8 hours; maximum dose: 3.2 g/day. Some experts generally recommend a maximum dose of 2.4 g/day for chronic use, except during a disease flare when up to 3.2 g/day may be considered for several weeks until flare resolves (Ref).

Dysmenorrhea

Dysmenorrhea:

Oral: Initial: 400 mg every 4 hours as needed or 600 to 800 mg every 6 to 8 hours as needed; maximum dose: 3.2 g/day. Begin at menses onset or 1 to 2 days prior to onset of menses for severe symptoms; usual duration: 1 to 5 days (Ref).

Fever

Fever (alternative agent):

IV, Oral: 200 to 400 mg every 4 to 6 hours as needed; if fever persists, may titrate up to 600 to 800 mg every 6 hours as needed; maximum dose: 3.2 g/day (Ref).

OTC labeling (patient-guided therapy): **Oral:** 200 mg every 4 to 6 hours as needed; if no relief, may increase to 400 mg every 4 to 6 hours as needed; maximum dose: 1.2 g/day. Use for >3 days is not recommended unless directed by health care provider.

Gout, treatment, acute flares

Gout, treatment, acute flares (off-label use):

Oral: Initial: 800 mg every 8 hours within 24 to 48 hours of flare onset; reduce dose as symptoms improve. Discontinue a few days after resolution of clinical signs (usual total duration: 5 to 7 days); longer duration may be required if therapy is delayed (Ref).

Migraine, acute treatment

Migraine, acute treatment:

Note: Limit use to ≤ 14 days per month to avoid medication-overuse headache (Ref). For use as monotherapy in mild to moderate attacks not associated with vomiting or severe nausea; may be used in combination with triptans for severe migraine (Ref). Administration early in the course of a migraine attack, at the first sign of pain, may improve response to treatment (Ref).

Oral: 400 to 600 mg once (Ref).

OTC labeling (patient-guided therapy): **Oral:** 400 mg at onset of symptoms.

Pain

Pain (monotherapy or as an adjunctive agent):

IV, Oral: 200 to 400 mg every 4 to 6 hours as needed or 600 to 800 mg every 6 to 8 hours as needed; maximum dose: 3.2 g/day (Ref). For postoperative pain, doses may be scheduled initially (Ref). Some experts suggest a usual maximum of 2.4 g/day for chronic use due to increased adverse effects with higher doses (Ref).

OTC labeling (patient-guided therapy): **Oral:** 200 mg every 4 to 6 hours as needed; if no relief, may increase to 400 mg every 4 to 6 hours as needed; maximum dose: 1.2 g/day. Use for >10 days is not recommended unless directed by health care provider.

Pericarditis, acute or recurrent

Pericarditis, acute or recurrent (off-label use):

Note: In patients with an indication for aspirin (eg, coronary artery disease), aspirin is generally preferred over other nonsteroidal anti-inflammatory drugs (NSAIDs). Avoid nonaspirin NSAIDs in patients with pericarditis secondary to acute myocardial infarction given lack of benefit and potential harm (Ref).

Oral: Initial: 600 to 800 mg every 8 hours or 600 mg every 6 hours until resolution of symptoms for at least 24 hours and normalization of inflammatory biomarkers (eg, C-reactive protein) if monitored; initial therapy typically lasts for ≥ 1 to 2 weeks. Gradually taper over several weeks by decreasing each dose by 200 to 400 mg every 1 to 2 weeks; during taper, ensure patient remains asymptomatic and inflammatory biomarkers remain normal (if monitored). Use in combination with colchicine. In patients at risk of NSAID-related GI toxicity, prophylaxis (generally with a proton pump inhibitor) is recommended (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function: IV, Oral:

CrCl ≥ 60 mL/minute: No dosage adjustment necessary (Ref).

CrCl >30 to <60 mL/minute: No dosage adjustment necessary (Ref). Use of analgesics other than nonsteroidal anti-inflammatory drugs may be preferred. If necessary, use the lowest effective dose for the shortest duration possible; avoid in patients at high risk for acute kidney injury (ie, volume depleted, hypotensive, elderly, or taking concurrent nephrotoxic medications) (Ref).

CrCl ≤ 30 mL/minute: Avoid use due to increased risk of acute kidney injury (Ref).

Hemodialysis, intermittent (thrice weekly): IV, Oral: Not significantly dialyzable (Ref): No dosage adjustment necessary. Avoid use in patients with residual kidney function (Ref).

Peritoneal dialysis: IV, Oral: No dosage adjustment necessary. Avoid use in patients with residual kidney function (Ref).

CRRT: IV, Oral: Avoid use (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): IV, Oral: Avoid use (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Nonsteroidal anti-inflammatory agents inhibit the prostaglandins required in cirrhosis to mitigate the effects of the renin-angiotensin-aldosterone system on the renal vasculature and cyclooxygenases in the gastrointestinal tract. Ibuprofen can precipitate renal insufficiency and/or increase the risk of upper gastrointestinal bleeding in the setting of cirrhosis (Ref).

Liver impairment prior to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: Avoid use (Ref).

Dosing: Older Adult

Refer to adult dosing. Consider reduced initial dosage. Unless alternative agents are ineffective and a gastroprotective agent can be administered, avoid short-term scheduled use in combination with corticosteroids, anticoagulants, or antiplatelet agents or chronic use with or without medications that increase risk for bleeding (Ref).

Dosing: Pediatric

(For additional information see "Ibuprofen: Pediatric drug information")

Dosage guidance:

Safety: To reduce the risk of adverse cardiovascular and GI effects, use the lowest effective dose for the shortest period of time to achieve treatment goals.

Dosage form information: Oral liquid products are available in 2 concentrations (concentrated infant drops: 50 mg/1.25 mL [40 mg/mL] and suspension: 100 mg/5 mL [20 mg/mL]); precautions should be taken to verify and avoid confusion between the different concentrations; dose should be clearly presented as "mg".

Pain

Pain (monotherapy or as an adjunctive agent):

IV: Ibuprofen injection (Caldolor): **Note:** Patients should be well hydrated prior to administration.

Infants ≥3 months to <6 months: IV: 10 mg/kg/dose as a single dose; maximum dose: 100 mg/dose.

Infants ≥6 months and Children <12 years: IV: 10 mg/kg/dose (maximum dose: 400 mg/dose) every 4 to 6 hours as needed; maximum daily dose: 40 mg/kg/day or 2,400 mg/day, whichever is less.

Children ≥12 years and Adolescents ≤17 years: IV: 400 mg every 4 to 6 hours as needed; maximum daily dose: 40 mg/kg/day or 2,400 mg/day, whichever is less.

Adolescents ≥ 18 years: IV: 400 to 800 mg every 6 hours as needed; maximum daily dose: 3,200 mg/**day**.

Oral:

Weight-directed dosing: Infants, Children, and Adolescents: Limited data available in infants < 6 months: Oral: 4 to 10 mg/kg/dose (maximum dose: 600 mg/dose) every 6 to 8 hours; maximum daily dose: 40 mg/kg/**day** or 2,400 mg/**day**, whichever is less (Ref).

Fixed dosing:

Infants ≥ 6 months and Children ≤ 11 years: Oral: See table based upon manufacturer's labeling; use of weight to select dose is preferred; if weight is not available, then use age; doses may be repeated every 6 to 8 hours; maximum: 4 doses/**day**; treatment of sore throat for > 2 days or use in infants and children < 3 years of age with sore throat is not recommended, unless directed by health care provider.

Ibuprofen Dosing

Weight (preferred)^a

Age

Dosage

(mg)

kg

lbs

5.4 to 8.1

12 to 17

6 to 11 months

50

8.2 to 10.8

18 to 23

12 to 23 months

75 to 80

10.9 to 16.3

24 to 35

2 to 3 years

100

16.4 to 21.7

36 to 47

4 to 5 years

150

21.8 to 27.2

48 to 59

6 to 8 years

200

27.3 to 32.6

60 to 71

9 to 10 years

200 to 250

32.7 to 43.2

72 to 95

11 years

300

^a Manufacturer's recommendations are based on weight in pounds (OTC labeling); weight in kg listed here is derived from pounds and rounded; kg weight listed also is adjusted to allow for continuous weight ranges in kg.

Children ≥ 12 years and Adolescents: Oral: 200 to 400 mg every 4 to 6 hours as needed; treatment of pain for >10 days is not recommended, unless directed by health care provider (Ref).

Cystic fibrosis, mild disease

Cystic fibrosis, mild disease (to slow lung disease progression): Limited data available: Children and Adolescents 6 to 17 years with $FEV_1 > 60\%$ predicted (Ref): Oral: Initial: 20 to 30 mg/kg/dose twice daily; titrate to achieve peak plasma concentrations of 50 to 100 mcg/mL; should not eat or take pancreatic enzymes for 2 hours after the ibuprofen dose. Dosing based on a study of 41 patients (ages: 5 to 39 years); mean required dose: ~ 25 mg/kg/dose twice daily; reported range: 16.2 to 31.6 mg/kg/dose every 12 hours required to achieve target concentration; results showed that chronic ibuprofen use (over 4 years) slowed the rate of decline in FEV_1 ; patients 5 to 13 years of age with mild lung disease were observed to have greatest benefit (Ref). A follow up observational study ($n=1,365$; ages: 6 to 17 years) under noncontrolled conditions (real world) showed significant improvement in the rate of decline of lung disease progression with chronic ibuprofen therapy (Ref).

Note: Timing of blood sampling postdose is based on dosage form: Oral suspension: Obtain blood samples at 30, 45, and 60 minutes postdose; tablets: Obtain blood samples at 1, 2, and 3 hours postdose (Ref).

Fever

Fever:

IV: Ibuprofen injection (CalAolor): **Note:** Patients should be well hydrated prior to administration.

Infants ≥ 3 months to <6 months: IV: 10 mg/kg/dose as a single dose; maximum dose: 100 mg/dose.

Infants ≥ 6 months and Children <12 years: IV: 10 mg/kg/dose (maximum dose: 400 mg/dose) every 4 to 6 hours as needed; maximum daily dose: 40 mg/kg/**day** or 2,400 mg/**day**, whichever is less.

Children ≥ 12 years and Adolescents ≤ 17 years: IV: 400 mg every 4 to 6 hours as needed; maximum daily dose: 40 mg/kg/**day** or 2,400 mg/**day**, whichever is less.

Adolescents ≥ 18 years: IV: Initial dose: 400 mg once, followed by 400 mg every 4 to 6 hours **or** 100 to 200 mg every 4 hours as needed; maximum daily dose: 3,200 mg/**day**.

Oral:

Weight-directed dosing: Infants ≥ 3 months weighing ≥ 5 kg, Children, and Adolescents: Limited data available in infants <6 months: Oral: 5 to 10 mg/kg/dose (maximum dose: 600 mg/dose) every 6 to 8 hours; maximum daily dose: 40 mg/kg/**day** or 2,400 mg/**day**, whichever is less (Ref).

Fixed dosing: **Note:** Treatment for >3 days is not recommended unless directed by health care provider.

Infants ≥ 6 months and Children ≤ 11 years: Oral: See table based upon manufacturer's labeling; use of weight to select dose is preferred; if weight is not available, then use age; doses may be repeated every 6 to 8 hours; maximum: 4 doses/**day**.

Ibuprofen Dosing

Weight (preferred)^a

Age

Dosage

(mg)

kg

lbs

5.4 to 8.1

12 to 17

6 to 11 months

50

8.2 to 10.8

18 to 23

12 to 23 months

75 to 80

10.9 to 16.3

24 to 35

2 to 3 years

100

16.4 to 21.7

36 to 47

4 to 5 years

150

21.8 to 27.2

48 to 59

6 to 8 years

200

27.3 to 32.6

60 to 71

9 to 10 years

200 to 250

32.7 to 43.2

72 to 95

11 years

300

^a Manufacturer's recommendations are based on weight in pounds (OTC labeling); weight in kg listed here is derived from pounds and rounded; kg weight listed also is adjusted to allow for continuous weight ranges in kg.

Children ≥ 12 years and Adolescents: Oral: 200 to 400 mg every 4 to 6 hours as needed (Ref).

Juvenile idiopathic arthritis

Juvenile idiopathic arthritis (JIA): Children and Adolescents: Oral: Usual range: 30 to 40 mg/kg/day in 3 to 4 divided doses; start at lower end of dosing range and titrate; patients with more severe disease may require up to 50 mg/kg/day; maximum dose: 800 mg/dose; maximum daily dose: 2,400 mg/day (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: Oral, IV (Caldolor): There are no dosage adjustments provided in the manufacturer's labeling; avoid use in advanced disease.

KDIGO guidelines provide the following recommendations for NSAIDs (Ref):

eGFR 30 to <60 mL/minute/1.73 m²: Avoid use in patients with intercurrent disease that increases risk of acute kidney injury.

eGFR <30 mL/minute/1.73 m²: Avoid use.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; use caution and discontinue if hepatic function worsens.

Adverse Reactions (Significant): Considerations

Cardiovascular effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs) is associated with an increased risk of serious adverse cardiovascular (CV) events, including **acute myocardial infarction (MI)**, **cerebrovascular accident**, and CV death. New-onset **hypertension** or **exacerbation of hypertension** may occur with NSAID use which may also contribute to an increased risk of CV events (Ref). New-onset or exacerbation of **heart failure** may also occur with cyclooxygenase (COX) NSAIDs (ie, coxibs) and nonselective NSAIDs, including ibuprofen, resulting in an increased risk of hospitalizations for heart failure and death in patients with heart failure (Ref).

Data collected by the Coxib and traditional NSAID Trialists' (CNT) Collaborative have shown that high-dose ibuprofen (2,400 mg daily) and diclofenac are associated with a similar CV risk as compared to coxibs and that naproxen may have the most favorable CV risk profile among NSAIDs analyzed (Ref); however, data from the PRECISION trial showed no difference with regards to risk between naproxen, ibuprofen, or celecoxib after a treatment duration of therapy of ~ 3 years (Ref). The FDA states that there are insufficient data to determine if risk of MI or stroke is definitely higher or lower for any particular NSAID as compared to another (Ref).

Mechanism: Dose- and time-related; inhibition of COX-2 by NSAIDs results in a reduction in the production of prostaglandin I₂ (prostacyclin) in the vascular endothelium (Ref); animal studies have shown that reduced prostacyclin activity may result in a predisposition to vascular injury (Ref). In addition, prostaglandins inhibit sodium resorption in the thick ascending loop of Henle and collecting tubule; therefore, a reduction in prostaglandin synthesis by NSAIDs may cause sodium and fluid retention and result in hypertension and decreased efficacy of diuretics (Ref).

Onset: Varied; increased risk may be apparent within the first weeks following initiation of treatment (Ref); longer duration of therapy may further increase risk (Ref).

Risk factors:

- ≥ 65 years of age
- Higher doses (especially with regards to CV thrombotic risk (Ref))
- Longer duration of use and frequent use (eg, ≥ 22 days per month (Ref))

- Preexisting cardiovascular disease (CVD) or presence of risk factors for CVD, including use following coronary artery bypass graft surgery (Ref)

- **Note:** Relative risk appears to be similar in those with and without known CVD or risk factors for CVD; however, absolute incidence of serious CV thrombotic events appears to be higher in patients with known CVD or risk factors for CVD due to an increased baseline risk (Ref)

Gastrointestinal events

Use of nonsteroidal anti-inflammatory drugs (NSAIDs), especially nonselective NSAIDs, such as ibuprofen, is associated with an increased risk of serious GI adverse events, including **gastrointestinal inflammation, gastrointestinal hemorrhage, gastrointestinal ulcer, and gastrointestinal perforation**; severity may range from asymptomatic to fatal (Ref).

Mechanism: Dose- and time-related; inhibition of cyclooxygenase (COX)-1 by NSAIDs results in a reduction in the production of mucosal-protective prostaglandin E₂ (Ref).

Onset: Varied; GI events can occur at any time during use and without warning symptoms. A longer duration of use (eg, ≥ 7 days (Ref)) is associated with a greater risk.

Risk factors:

- ≥ 65 years of age (Ref)
- Longer duration of use (eg, ≥ 7 days (Ref))
- Higher doses (Ref)
- Prior history of peptic ulcer disease and/or GI bleeding (Ref)
- Concomitant use of agents known to increase the risk of GI bleeding (eg, aspirin (Ref), anticoagulants, corticosteroids (Ref), selective serotonin reuptake inhibitors (Ref))
- Comorbid *Helicobacter pylori* infection (Ref)
- Advanced liver disease/cirrhosis
- Coagulopathy
- Smoking
- Consumption of alcohol
- People with poor general health status
- Small intestine damage: Small intestine bacterial overgrowth (SIBO), including SIBO induced by proton pump inhibitor therapy, may be associated with an increased risk of small intestine damage (Ref)

Hematologic effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs), including ibuprofen, is associated with **prolonged bleeding time** and an increased risk for **hemorrhage** (Ref).

In addition, drug-induced **hemolytic anemia** may occur (Ref). Rarely, NSAID use has been associated with potentially severe blood dyscrasias (eg, **agranulocytosis, aplastic anemia, neutropenia, thrombocytopenia**) (Ref).

Mechanism:

Prolonged bleeding time: Inhibition of cyclooxygenase (COX)-1 by nonselective NSAIDs causes a decrease in the production of prostaglandins, prostacyclins, and thromboxanes, including thromboxane A₂ (TxA₂) (Ref). As a result, patients may exhibit a decrease in platelet adhesion and aggregation and subsequent prolonged bleeding time (Ref).

Blood dyscrasias: Not clearly established; anemia may be due to occult or gross blood loss, fluid retention, or an incompletely described effect on erythropoiesis.

Onset:

Prolonged bleeding time: Rapid; suppression of platelet COX-1 activity occurs within hours of administration (Ref). In patients receiving antithrombotic therapy after myocardial infarction, the use of NSAIDs has been associated with an increased risk of bleeding and excess thrombotic events, even after short-term treatment (eg, <3 days) (Ref).

Risk factors:

- Bleeding events:
- Preexisting coagulation disorders
- Concomitant use of agents known to increase the risk of bleeding (eg, anticoagulants (Ref), antithrombotics (Ref), antiplatelet agents [eg, aspirin], selective serotonin reuptake inhibitors (Ref), or serotonin norepinephrine reuptake inhibitors)
- Use during and immediately following surgical procedures (Ref)

Hepatic effects

Nonsteroidal anti-inflammatory drugs (NSAIDs), including ibuprofen, may cause mild transaminase elevations, especially with higher doses. Rarely, serious liver injury may occur (Ref). Cholestatic and mixed patterns of injury have been reported. Hepatic effects may occur following severe hypersensitivity reactions (eg, toxic epidermal necrosis, Stevens-Johnson syndrome) and are characterized by immune-mediated symptoms (eg, fever rash, eosinophilia, lymphadenopathy) (Ref). Severe liver injury requiring liver transplantation has also been reported (Ref). Most cases of liver injury are likely reversible following discontinuation; full recovery may take several months (Ref). However, chronic vanishing bile duct syndrome with chronic liver failure has been reported following cholestatic liver injury (Ref).

Mechanism: Not clearly established; dose-related has been suggested (Ref). Proposed mechanisms include a toxic metabolite or a hypersensitivity reaction (Ref).

Onset: Varied; onset of NSAID-induced hepatotoxicity is generally classified as moderate (30 to 90 days) to long (>90 days) (Ref). For ibuprofen, a mean time of 12 days (range: 1 to 42 days) was reported in one study (Ref).

Risk factors:

- Higher doses (Ref)
- Prior NSAID-related liver injury (Ref)
- **Note:** Cross-reactivity may occur among propionic acid derivatives (eg, ibuprofen, naproxen, ketoprofen) (Ref)

Hypersensitivity reactions (immediate and delayed)

Hypersensitivity reactions (immediate and delayed) involving the skin (eg, **angioedema, urticaria**), airways (eg, dyspnea, rhinorrhea), and/or other organs have been reported (Ref). Clinical phenotypes of nonsteroidal anti-inflammatory drug (NSAID) hypersensitivity reactions include NSAID-exacerbated respiratory disease (NERD), NSAID-induced urticaria/angioedema (NIUA), NSAID-exacerbated cutaneous disease (NECD), and single NSAID-induced urticaria/angioedema or anaphylaxis (Ref). Delayed hypersensitivity reactions ranging from mild reactions, such as **fixed drug eruption** (Ref), to more serious reactions, including acute generalized exanthematous pustulosis (AGEP), drug rash with eosinophilia and systemic symptoms (DRESS), and **Stevens-Johnson syndrome** (SJS), have also been associated with ibuprofen (Ref).

Mechanism:

Immediate reactions: Non-dose-related; most reactions (ie, NERD, NECD, NIUA) are non-immunologic related to inhibition of cyclooxygenase-1 (COX-1) with subsequent activation of mast cells and eosinophils causing release of inflammatory mediators including cysteinyl-leukotrienes (cysLTs) (Ref). Some immediate reactions are IgE-mediated (Ref).

Delayed reactions: Delayed hypersensitivity reactions are T-cell-mediated (Ref).

Onset:

Immediate reactions: Rapid; occur within 1 hour of administration but may occur several hours after exposure (Ref).

Delayed reactions (including DRESS, AGEP, and SJS): Varied, generally occurs after 1 to 8 weeks after initiation (Ref), although some patients may develop symptoms within 24 hours (Ref).

Risk factors:

- Presence of chronic rhinosinusitis with nasal polyps, family history of NERD, and/or severe asthma may increase the risk of NERD (Ref). The prevalence of NERD in adult patients with asthma is ~10% to 20% (Ref).
- Chronic urticaria increases the risk of NECD (Ref). NSAID-induced reactions are less frequent and less intense when chronic urticaria is in remission or under control (Ref). Approximately 12% to 30% of patients with chronic idiopathic urticaria develop exacerbations of their disease with use of ibuprofen and other COX-1 inhibitors (Ref).
- Cross-reactivity between aspirin and NSAIDs, including ibuprofen (with predominant COX-1 inhibition) have been described in patients with a history of NERD, NECD, and NIUA (Ref). Cross-reactivity between aspirin/NSAID and acetaminophen, a weak COX inhibitor, and between aspirin/NSAID and nonselective COX-2 inhibitors (eg, meloxicam, nimesulide) may occur (Ref). Although selective COX-2 inhibitors (eg, celecoxib, etoricoxib) are generally tolerated in patients with NERD (Ref), cross-reactions may occur, especially in patients with histories of urticaria/angioedema (Ref).
- Cross-reactivity may occur among propionic acid derivatives (eg, ibuprofen, naproxen, ketoprofen) in patients with histories of immediate hypersensitivity reactions to ibuprofen but tolerance to aspirin (Ref)

Kidney effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs), including ibuprofen, is associated with an increased risk of several kidney-specific effects: **Hemodynamically-mediated acute kidney injury, interstitial nephritis** (with or without nephrotic syndrome), and **renal papillary necrosis** in all ages.

Hemodynamically-mediated acute kidney injury (AKI): Hemodynamically-mediated AKI may occur following use of either cyclooxygenase (COX)-2 selective NSAIDs (ie, coxibs) or nonselective NSAIDs, including ibuprofen (Ref); the risk may be greater with nonselective NSAIDs, especially indomethacin (Ref). The risk of developing AKI is decreased upon discontinuation (Ref). In patients who develop AKI, kidney function is likely to return to baseline following prompt discontinuation of the offending NSAID and supportive care (Ref); however, the mechanism of the damage and other concurrent factors can contribute to irreversibility.

Acute interstitial nephritis (AIN) with or without nephrotic syndrome: Patients may develop NSAID-associated proteinuria combined with interstitial nephritis and varying degrees of kidney impairment; the “classic triad” of fever, rash, and eosinophilia is less commonly observed in NSAID-associated AIN than with antibiotic-induced AIN (Ref). Kidney histology may reveal minimal change glomerulonephritis or membranous nephropathy (Ref). While use of ibuprofen has been associated with this clinical picture, the risk may be greatest with fenoprofen as compared to other NSAIDs (Ref). Proteinuria generally improves within weeks following discontinuation; full recovery may require treatment and take up to a year (Ref).

Papillary necrosis: Chronic use of NSAIDs, including ibuprofen, has resulted in the development of papillary necrosis, which may occur in conjunction with chronic interstitial nephritis and progressive decline in glomerular filtration rate as a clinical syndrome known as analgesic nephropathy (Ref). However, controversy exists on the degree to which NSAID use increases the risk for chronic kidney disease and analgesic nephropathy (Ref). Acute papillary necrosis may occur following NSAID overdose, especially in a setting of severe dehydration or intravascular volume depletion (Ref).

Mechanism:

Hemodynamically-mediated AKI: Dose- and time-related; inhibition of cyclooxygenase (COX)-1 and COX-2 by NSAIDs results in a reduced production of nephroprotective prostaglandins and subsequent attenuation of renal vasodilation (Ref). In addition, an increase in vasoconstriction of the afferent arteriole and impaired renal blood flow causes a reduction in the glomerular capillary pressure and filtration (Ref).

AIN with or without nephrotic syndrome: Not clearly established. Following inhibition of COX-1 and COX-2 by NSAIDs, arachidonic acid is formed which may be further metabolized to leukotrienes via the lipoxygenase pathway; leukotrienes may increase vascular permeability within glomerular capillaries and peritubular capillaries and increase lymphocyte recruitment and activation (Ref).

Papillary necrosis: Time-related; exact mechanism is not clearly established; may be due to direct toxicity and/or inhibition of prostaglandin-mediated vasodilation resulting in ischemic necrosis (Ref).

Onset:

AKI: Rapid; may occur within days of treatment initiation (Ref).

AIN with or without nephrotic syndrome: Varied; mean time of onset of ~5 months (range: 2 weeks to 18 months) has been described (Ref).

Risk factors:

- AKI:
 - Preexisting kidney impairment
 - Chronic kidney disease
 - **Note:** High cumulative doses (eg, ibuprofen >700 mg/day) may increase the risk for progression of chronic kidney disease (Ref)
 - ≥65 years of age (Ref)
 - **Note:** NSAID-associated AKI may also occur in pediatric patients, even at therapeutic doses (Ref)
- Hemodynamically-mediated AKI:
 - Preexisting conditions which result in decreased effective arterial circulation (ie, conditions where renal blood flow/renal perfusion may be dependent on prostaglandin-mediated vasodilation) (Ref):

Volume depletion (eg, due to concomitant diuretic use, nausea, vomiting)

Heart failure (Ref)

Cirrhosis and ascites (Ref)

Nephrotic syndrome

- Concomitant use of diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or calcineurin inhibitors (Ref)
- AIN with or without nephrotic syndrome: Prior history of NSAID-induced nephrotic syndrome; recurrence has been described (Ref)
- Papillary necrosis (acute):
 - Massive NSAID ingestion (Ref)
 - Dehydration (Ref)
 - Intravascular volume depletion (Ref)
 - Papillary necrosis (chronic)/analgesic nephropathy: Chronic concomitant use of other analgesics (eg, aspirin, acetaminophen) (Ref)

Pulmonary hypertension

Pulmonary hypertension has occurred following early (prophylactic) administration and treatment of patent ductus arteriosus (PDA) in preterm or low birth weight neonates; cases have been reported

with both tromethamine ibuprofen (not available in the United States) and L-lysine ibuprofen therapy (Ref).

Mechanism: Unknown; several hypotheses have been proposed. The first relates to timing of treatment in which the administration of drug and PDA closure occurs before the normal decrease in pulmonary vascular resistance has occurred (Ref). Another hypothesis involves the effects of the acidic pH of the ibuprofen solution, namely tromethamine buffered solutions, which may lead to precipitation and microembolism of the lung (Ref).

Onset: Variable; has occurred within the first hour following the administration of the first or second dose of treatment (Ref).

Risk factors:

- Lower gestational age (Ref)
- Lower birthweight (less than third percentile for age) (Ref)
- Preexisting pulmonary hypertension (Ref)
- Early administration of treatment (<6 hours postnatal age in premature neonates) (Ref)
- Maternal hypertension of pregnancy (Ref)
- Maternal oligohydramnios (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

Oral:

>10%: Hematologic & oncologic: Decreased hemoglobin (17% to 23%)

1% to 10%:

Cardiovascular: Edema (1% to 3%)

Dermatologic: Maculopapular rash, pruritus (1% to 3%), skin rash (3% to 9%)

Endocrine & metabolic: Fluid retention (1% to 3%)

Gastrointestinal: Abdominal cramps (1% to 3%), abdominal distress (1% to 3%), abdominal pain (1% to 3%), bloating (1% to 3%), constipation (1% to 3%), decreased appetite (1% to 3%), diarrhea (1% to 3%), dyspepsia (1% to 3%), epigastric pain (3% to 9%), flatulence (1% to 3%), heartburn (3% to 9%), nausea (3% to 9%), nausea and vomiting (1% to 3%)

Nervous system: Dizziness (3% to 9%), headache (1% to 3%), nervousness (1% to 3%)

Otic: Tinnitus (1% to 3%)

<1%:

Cardiovascular: Cardiac arrhythmia, increased blood pressure, palpitations, sinus bradycardia, sinus tachycardia

Dermatologic: Alopecia, erythema multiforme, skin photosensitivity, urticaria, vesiculobullous dermatitis

Endocrine & metabolic: Acidosis, gynecomastia, heavy menstrual bleeding, hypoglycemia

Gastrointestinal: Duodenal ulcer, gastric ulcer, gastritis, gingival ulceration, melena, pancreatitis, xerostomia

Genitourinary: Azotemia, cystitis, hematuria

Hematologic & oncologic: Agranulocytosis, aplastic anemia, decreased hematocrit, eosinophilia, hemolytic anemia, hemorrhage, Henoch-Schonlein purpura, neutropenia, thrombocytopenia, ulcer with hemorrhage

Hepatic: Abnormal hepatic function tests, hepatitis, jaundice

Hypersensitivity: Angioedema, serum sickness

Nervous system: Abnormal dreams, chills, confusion, depression, drowsiness, emotional lability, hallucination, idiopathic intracranial hypertension, insomnia, paresthesia

Neuromuscular & skeletal: Systemic lupus erythematosus

Ophthalmic: Amblyopia, cataract, conjunctivitis, diplopia, optic neuritis, xerophthalmia

Otic: Hearing loss

Renal: Decreased creatinine clearance, polyuria, renal papillary necrosis

Respiratory: Bronchospasm, epistaxis, rhinitis

Miscellaneous: Fever

Injection: Ibuprofen (Calabolor):

>10%:

Endocrine & metabolic: Hypokalemia (4% to 19%)

Gastrointestinal: Flatulence (16%), vomiting (22%; children: $\geq 2\%$)

Hematologic & oncologic: Anemia (4% to 36%; children: $\geq 2\%$), eosinophilia (26%), hypoproteinemia (10% to 13%), neutropenia (7% to 13%) (table 1) Neutropenia

Table 1: **Ibuprofen: Adverse Reaction: Ne**

Drug (Ibuprofen)	Placebo	Population	Dose	Dosage Form	Indication
13%	7%	Adults	400 mg every 4 hours for 24 hours	IV	Fever
7%	7%	Adults	200 mg every 4 hours for 24 hours	IV	Fever
7%	7%	Adults	100 mg every 4 hours for 24 hours	IV	Fever

Infection: Bacteremia (13%)

Nervous system: Headache (12%; children: $\geq 2\%$)

1% to 10%:

Cardiovascular: Hypertension ($\leq 10\%$) (table 2) Hypertension, hypotension (7% to 10%), peripheral edema (3%)

Table 2: **Ibuprofen: Adverse Reaction: Hyp**

Drug (Ibuprofen)	Placebo	Population	Dose	Dosage Form	Indication
10%	0%	Adults	400 mg every 4 hours for 24 hours	IV	Fever
0%	0%	Adults	200 mg every 4 hours for 24 hours	IV	Fever
0%	0%	Adults	100 mg every 4 hours for 24 hours	IV	Fever

Endocrine & metabolic: Hypernatremia (7% to 10%), hypoalbuminemia (10%), increased lactate dehydrogenase (7% to 10%)

Gastrointestinal: Abdominal distress ($\leq 3\%$), diarrhea (10%), dyspepsia (1% to 4%), nausea (children: $\geq 2\%$)

Genitourinary: Urinary retention (5%)

Hematologic & oncologic: Hemorrhage (10%), thrombocytopenia (3% to 10%), wound hemorrhage (3%)

Local: Infusion-site pain (children: $\geq 2\%$)

Nervous system: Dizziness (4% to 6%)

Renal: Increased blood urea nitrogen (10%)

Respiratory: Bacterial pneumonia (3% to 10%), cough (3%)

Injection: Ibuprofen lysine (NeoProfen) (as reported in premature infants):

>10%:

Dermatologic: Skin irritation ($\leq 16\%$), skin lesion ($\leq 16\%$)

Endocrine & metabolic: Hypocalcemia (12%), hypoglycemia (12%)

Gastrointestinal: Enterocolitis (22%)

Hematologic & oncologic: Anemia (32%), hemorrhage (32%; primarily intraventricular)

Infection: Sepsis (43%)

Nervous system: Intraventricular hemorrhage (29%)

Respiratory: Apnea (28%), respiratory tract infection (19%)

1% to 10%:

Cardiovascular: Edema (4%)

Endocrine & metabolic: Adrenocortical insufficiency (7%), hypernatremia (7%)

Genitourinary: Decreased urine output (3%), urinary tract infection (9%)

Renal: Increased blood urea nitrogen (7%), increased serum creatinine (3%), renal insufficiency (6%)

Respiratory: Atelectasis (4%), respiratory failure (10%)

Frequency not defined (any formulation):

Cardiovascular: Hypotension, tachycardia

Endocrine & metabolic: Hyperglycemia

Gastrointestinal: Abdominal distention, cholestasis, gastritis, gastroesophageal reflux disease, intestinal obstruction

Hematologic & oncologic: Neutropenia, prolonged bleeding time

Hepatic: Jaundice

Infection: Infection

Local: Injection-site reaction

Nervous system: Seizure

Miscellaneous: Reduced intake of food/fluids

Postmarketing (any formulation):

Cardiovascular: Acute myocardial infarction (Ref), exacerbation of hypertension (Ref), heart failure (Ref), thrombosis

Dermatologic: Exfoliative dermatitis, fixed drug eruption (Ref), skin rash, Stevens-Johnson syndrome (Ref), toxic epidermal necrolysis (Ref)

Gastrointestinal: Gastrointestinal hemorrhage (Ref), gastrointestinal inflammation, gastrointestinal perforation (including esophageal perforation, intestinal, stomach) (Ref), gastrointestinal ulcer (Ref), necrotizing enterocolitis

Genitourinary: Oliguria (Ref)

Hematologic & oncologic: Thrombocytopenia (Ref)

Hepatic: Hepatotoxicity (idiosyncratic) (Ref), increased serum alanine aminotransferase, increased serum aspartate aminotransferase

Hypersensitivity: Anaphylaxis (Ref), drug reaction with eosinophilia and systemic symptoms syndrome (Ref), hypersensitivity reaction

Nervous system: Aseptic meningitis (Ref), cerebrovascular accident (Ref)

Ophthalmic: Blurred vision, scotoma, vision color changes, visual disturbance

Renal: Acute kidney injury (Ref), interstitial nephritis (Ref), kidney failure (Ref)

Respiratory: Pulmonary hypertension (Ref)

Miscellaneous: Radiation recall phenomenon (dermatitis) (Ref)

Contraindications

Hypersensitivity to ibuprofen (eg, anaphylactic reactions, serious skin reactions) or any component of the formulation; history of asthma, urticaria, or allergic-type reaction to aspirin or other NSAIDs; aspirin triad (eg, bronchial asthma, aspirin intolerance, rhinitis); use in the setting of coronary artery bypass graft (CABG) surgery

Ibuprofen lysine (NeoProfen): Proven or suspected infection that is untreated; congenital heart disease in whom patency of the PDA is necessary for satisfactory pulmonary or systemic blood flow (eg, pulmonary atresia, severe coarctation of the aorta, severe tetralogy of Fallot); bleeding (especially those with active intracranial hemorrhage or GI bleeding); thrombocytopenia; coagulation defects; proven or suspected necrotizing enterocolitis; or significant renal function impairment.

Canadian labeling: Additional contraindications (not in US labeling): Cerebrovascular bleeding or other bleeding disorders; active gastric/duodenal/peptic ulcer, active GI bleeding; inflammatory bowel disease; uncontrolled heart failure; moderate [IV formulation only] to severe renal impairment (creatinine clearance [CrCl] <30 mL/minute); deteriorating renal disease; moderate [IV formulation only] to severe hepatic impairment; active hepatic disease; hyperkalemia; third trimester of pregnancy; breast-feeding; patients <18 years of age [IV formulation only]; patients <12 years of age [oral formulation only]; systemic lupus erythematosus [oral formulation only].

OTC labeling: When used for self-medication, do not use if previous allergic reaction to any other pain reliever/fever reducer; prior to or following cardiac surgery.

Warnings/Precautions

Concerns related to adverse effects:

- **CNS effects:** May cause drowsiness, dizziness, blurred vision, and other neurologic effects which may impair physical or mental abilities; patients must be cautioned about performing tasks which require mental alertness (eg, operating machinery or driving).
- **Hyperkalemia:** Nonsteroidal anti-inflammatory drug (NSAID) use may increase the risk of hyperkalemia, particularly in patients ≥ 65 years of age, in patients with diabetes or renal disease, and with concomitant use of other agents capable of inducing hyperkalemia (eg, ACE inhibitors). Monitor potassium closely.
- **Ophthalmic events:** Blurred/diminished vision, scotomata, and changes in color vision have been reported. Discontinue therapy and refer for ophthalmologic evaluation if symptoms occur. Periodically evaluate vision in all patients receiving long-term therapy.

Disease-related concerns:

- **Aseptic meningitis:** May increase the risk of aseptic meningitis, especially in patients with systemic lupus erythematosus and mixed connective tissue disorders.
- **Asthma:** Contraindicated in patients with aspirin-sensitive asthma; severe and potentially fatal bronchospasm may occur. Use caution in patients with other forms of asthma.

- Bariatric surgery: Gastric ulceration: Avoid chronic use of oral nonselective NSAIDs after bariatric surgery; development of anastomotic ulcerations/perforations may occur (Bhangu 2014; Mechanick 2020). Short-term use of celecoxib or IV ketorolac is recommended as part of a multimodal pain management strategy for postoperative pain (Chou 2016b; Horsley 2019; Thorell 2016).
- Hepatic impairment: Use with caution in patients with hepatic impairment.
- Renal impairment: Use with caution in patients with renal impairment. Use of ibuprofen lysine (NeoProfen) is contraindicated in preterm infants with significant renal impairment.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [“Inactive” 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol derivative with caution in neonates. See manufacturer’s labeling.
- Ibuprofen injection (Calaolor): Must be diluted prior to administration; hemolysis can occur if not diluted.
- Ibuprofen lysine injection (NeoProfen): May alter signs of infection. May inhibit platelet aggregation; monitor for signs of bleeding. May displace bilirubin; use caution when total bilirubin is elevated. Long-term evaluations of neurodevelopment, growth, or diseases associated with prematurity following treatment have not been conducted. Avoid extravasation.
- Phenylalanine: Some products may contain phenylalanine.
- Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer’s labeling.
- Propylene glycol: Some dosage forms may contain propylene glycol; large amounts are potentially toxic and have been associated hyperosmolality, lactic acidosis, seizures and respiratory depression; use caution (AAP [“Inactive” 1997]; Zar 2007).

Other warnings/precautions:

- Self-medication (OTC use): Prior to self-medication, patients should contact health care provider if they have had recurring stomach pain or upset, ulcers, bleeding problems, high blood pressure, heart or kidney disease, other serious medical problems, are currently taking a diuretic, aspirin, anticoagulant, or are ≥ 60 years of age. If patients are using for migraines, they should also contact health care provider if they have not had a migraine diagnosis by health care provider, a headache that is different from usual migraine, worst headache of life, fever and neck stiffness, headache from head injury or coughing, first headache at ≥ 50 years of age, daily headache, or migraine requiring bed rest. Do not exceed recommended dosages due to an increased risk of GI bleeding. Stop use and consult a health care provider if symptoms do not improve within first 24 hours of use (children) get worse, or newly appear, fever lasts for >3 days or pain lasts >3 days (children) and >10 days (adults). Do not give for >10 days unless instructed by healthcare provider. Consuming ≥ 3 alcoholic beverages/day or taking longer than recommended may increase the risk of GI bleeding.
- Surgical/dental procedures: Withhold for at least 4 to 6 half-lives prior to surgical or dental procedures.

Warnings: Additional Pediatric Considerations

Oral liquid products are available in 2 concentrations (concentrated infant drops: 50 mg/1.25 mL [40 mg/mL] and suspension: 100 mg/5 mL [20 mg/mL]); precautions should be taken to verify and avoid confusion between the different concentrations; dose should be clearly presented as "mg".

IV ibuprofen is as effective as IV indomethacin for the treatment of patent ductus arteriosus (PDA) in preterm neonates, but is less likely to cause adverse effects on renal function (eg, oliguria, increased serum creatinine) (Aranda 2006; Lago 2002; Ohlsson 2013; Van Overmeire 2000). Ibuprofen (compared to indomethacin) also has been shown to decrease the risk of developing NEC (Ohlsson 2013).

Use with caution in neonates with controlled infection or those at risk for infection; ibuprofen may alter the usual signs of infection. Use with caution in neonates when total bilirubin is elevated; ibuprofen may displace bilirubin from albumin-binding sites.

Avoid extravasation of ibuprofen lysine injection (NeoProfen); IV solution may be irritating to tissues.

Some dosage forms may contain propylene glycol; in neonates, large amounts of propylene glycol delivered orally, intravenously (eg, >3,000 mg/day), or topically have been associated with potentially fatal toxicities which can include metabolic acidosis, seizures, renal failure, and CNS depression; toxicities have also been reported in children and adults including hyperosmolality, lactic acidosis, seizures, and respiratory depression; use caution (AAP 1997; Shehab 2009).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Capsule, Oral:

Aavil: 200 mg

Advil: 200 mg [contains coconut oil (copra/cocos nucifera oil)]

Aavil Liqui-Gels minis: 200 mg

Advil Migraine: 200 mg [contains fd&c red #40 (allura red ac dye), quinoline yellow (d&c yellow #10)]

FT Ibuprofen: 200 mg

FT Ibuprofen: 200 mg [gluten free; contains fd&c blue #1 (brilliant blue)]

FT Ibuprofen Minis: 200 mg

GoodSense Ibuprofen: 200 mg [DSC] [gluten free; contains fd&c blue #1 (brilliant blue)]

Motrin IB: 200 mg

Propinal: 200 mg

Generic: 200 mg

Solution, Intravenous [preservative free]:

Calalor: 800 mg/200 mL (200 mL); 800 mg/8 mL (8 mL)

Solution, Intravenous, as lysine [preservative free]:

NeoProfen: 10 mg/mL (2 mL)

Generic: 10 mg/mL (2 mL)

Suspension, Oral:

Childrens Advil: 100 mg/5 mL (120 mL) [fruit flavor]

Childrens Advil: 100 mg/5 mL (120 mL) [contains edetate (edta) disodium, fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, sodium benzoate]

Childrens Advil: 100 mg/5 mL (120 mL) [alcohol free; grape flavor]

Childrens Advil: 100 mg/5 mL (120 mL) [alcohol free; contains edetate (edta) disodium, fd&c blue #1 (brilliant blue), fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, sodium benzoate; grape flavor]

Childrens Advil: 100 mg/5 mL (120 mL [DSC]) [alcohol free; contains fd&c blue #1 (brilliant blue), propylene glycol, sodium benzoate]

Childrens Advil: 100 mg/5 mL (120 mL) [alcohol free, dye free; contains edetate (edta) disodium, polysorbate 80, propylene glycol, sodium benzoate]

Childrens Advil: 100 mg/5 mL (30 mL, 120 mL) [alcohol free, dye free; contains edetate (edta) disodium, polysorbate 80, propylene glycol, sodium benzoate; white grape flavor]

Childrens Advil: 100 mg/5 mL (120 mL) [alcohol free, dye free, sugar free; contains edetate (edta) disodium, polysorbate 80, propylene glycol, sodium benzoate; berry flavor]

Childrens Ibuprofen: 100 mg/5 mL (5 mL [DSC]) [contains fd&c red #40 (allura red ac dye), polysorbate 80, quinoline yellow (d&c yellow #10), sodium benzoate]

Childrens Ibuprofen: 100 mg/5 mL (5 mL [DSC]) [alcohol free, dye free; contains corn starch, polysorbate 80, sodium benzoate]

Childrens Ibuprofen: 100 mg/5 mL (5 mL); 200 mg/10 mL (10 mL) [alcohol free, dye free; contains polysorbate 80, sodium benzoate; unflavored flavor]

Childrens Motrin: 100 mg/5 mL (120 mL) [alcohol free; contains fd&c blue #1 (brill blue) aluminum lake, fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate]

Childrens Motrin: 100 mg/5 mL (30 mL, 120 mL) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, quinoline yellow (d&c yellow #10), sodium benzoate; berry flavor]

Childrens Motrin: 100 mg/5 mL (120 mL) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate]

Childrens Motrin: 100 mg/5 mL (120 mL, 240 mL) [alcohol free, dye free; contains polysorbate 80, sodium benzoate; berry flavor]

FT Ibuprofen Childrens: 100 mg/5 mL (118 mL) [contains fd&c blue #1 (brilliant blue), polysorbate 80, propylene glycol, sodium benzoate; grape flavor]

FT Ibuprofen Childrens: 100 mg/5 mL (118 mL, 237 mL) [contains fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, quinoline yellow (d&c yellow #10), sodium benzoate; berry flavor]

FT Ibuprofen Childrens: 100 mg/5 mL (118 mL) [contains fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, sodium benzoate; bubble-gum flavor]

FT Ibuprofen Childrens: 100 mg/5 mL (118 mL) [dye free; contains polysorbate 80, propylene glycol, sodium benzoate; berry flavor]

FT Ibuprofen Infants: 50 mg/1.25 mL (30 mL) [dye free, gluten free; contains polysorbate 80, sodium benzoate; berry flavor]

FT Ibuprofen Infants: 50 mg/1.25 mL (15 mL) [gluten free, paraben free; contains fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate; berry flavor]

GoodSense Ibuprofen Childrens: 100 mg/5 mL (120 mL) [alcohol free, dye free, gluten free; contains polysorbate 80, sodium benzoate, sorbitol; berry flavor]

Ibuprofen Childrens: 100 mg/5 mL (118 mL, 237 mL) [contains fd&c blue #1 (brilliant blue), polysorbate 80, propylene glycol, sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (118 mL) [contains fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, quinoline yellow (d&c yellow #10), sodium benzoate; berry flavor]

Ibuprofen Childrens: 100 mg/5 mL (118 mL, 237 mL) [contains fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (5 mL [DSC]) [alcohol free; contains corn starch, fd&c blue #1 (brilliant blue), fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate; grape flavor]

Ibuprofen Childrens: 100 mg/5 mL (118 mL [DSC]) [alcohol free; contains corn starch, fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (118 mL, 120 mL, 237 mL, 240 mL) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, propylene glycol, quinoline yellow (d&c yellow #10), sodium benzoate; berry flavor]

Ibuprofen Childrens: 100 mg/5 mL (120 mL [DSC], 240 mL [DSC]) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, quinoline yellow (d&c yellow #10), sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (240 mL [DSC]) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, quinoline yellow (d&c yellow #10), sodium benzoate; berry flavor]

Ibuprofen Childrens: 200 mg/10 mL (10 mL [DSC]) [alcohol free; contains corn starch, fd&c blue #1 (brilliant blue), fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate; grape flavor]

Ibuprofen Childrens: 100 mg/5 mL (118 mL [DSC]) [alcohol free, dye free; contains corn starch, polysorbate 80, sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (5 mL [DSC]) [alcohol free, dye free; contains polysorbate 80, propylene glycol, sodium benzoate; berry flavor]

Ibuprofen Childrens: 100 mg/5 mL (120 mL) [alcohol free, dye free; contains polysorbate 80, sodium benzoate; berry flavor]

Ibuprofen Childrens: 200 mg/10 mL (10 mL [DSC]) [alcohol free, dye free; contains polysorbate 80, propylene glycol, sodium benzoate; berry flavor]

Ibuprofen Childrens: 100 mg/5 mL (5 mL) [alcohol free, dye free, gluten free; contains fd&c red #40 (allura red ac dye), propylene glycol, quinoline yellow (d&c yellow #10), sodium benzoate]

Ibuprofen Childrens: 100 mg/5 mL (118 mL, 237 mL) [dye free; contains polysorbate 80, propylene glycol, sodium benzoate; berry flavor]

Infants Aadvil: 50 mg/1.25 mL (30 mL) [alcohol free, dye free; contains edetate (edta) disodium, polysorbate 80, propylene glycol, sodium benzoate]

Infants Advil: 50 mg/1.25 mL (15 mL) [alcohol free, dye free; contains edetate (edta) disodium, polysorbate 80, propylene glycol, sodium benzoate; white grape flavor]

Motrin Infants Drops: 50 mg/1.25 mL (15 mL) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate, sorbitol]

Motrin Infants Drops: 50 mg/1.25 mL (15 mL) [alcohol free; contains fd&c red #40 (allura red ac dye), polysorbate 80, sodium benzoate, sorbitol; berry flavor]

Motrin Infants Drops: 50 mg/1.25 mL (30 mL) [alcohol free, dye free; contains polysorbate 80, sodium benzoate, sorbitol]

Generic: 100 mg/5 mL (5 mL, 118 mL, 120 mL, 473 mL); 200 mg/10 mL (10 mL)

Tablet, Oral:

Aadvil: 200 mg [antihistamine free, caffeine free, salt free, sugar free; contains corn starch]

Advil: 200 mg

Advil: 200 mg [contains corn starch, methylparaben (methyl p-hydroxybenzoate), propylparaben (propyl p-hydroxybenzoate), sodium benzoate]

Advil Junior Strength: 100 mg

FT Ibuprofen: 200 mg [gluten free; contains corn starch]

FT Pain Relief: 200 mg [contains corn starch]

GoodSense Ibuprofen: 200 mg [gluten free; contains corn starch, fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake]

I-Prin: 200 mg [DSC] [contains corn starch]

IBU: 400 mg, 600 mg, 800 mg

Medi-First Ibuprofen: 200 mg [contains corn starch]

Motrin IB: 200 mg [contains corn starch, fd&c red #40(allura red ac)aluminum lake, fd&c yellow #6(sunset yellow)alumin lake]

Generic: 200 mg, 400 mg, 600 mg, 800 mg

Tablet Chewable, Oral:

Advil Junior Strength: 100 mg [scored; contains aspartame, fd&c blue #2 (indigo carm) aluminum lake; grape flavor]

FT Ibuprofen IB Childrens: 100 mg [scored; contains aspartame, fd&c yellow #6(sunset yellow)alumin lake, soybean oil; orange flavor]

Motrin Childrens: 100 mg [contains aspartame, fd&c blue #1 (brill blue) aluminum lake, soybean oil]

Motrin Childrens: 100 mg [dye free; contains aspartame, soybean oil]

Generic Equivalent Available: US

Yes

Pricing: US

Capsules (Advil Liqui-Gels minis Oral)

200 mg (per each): \$0.25

Capsules (Advil Migraine Oral)

200 mg (per each): \$0.17

Capsules (Advil Oral)

200 mg (per each): \$0.22

Capsules (Ibuprofen Oral)

200 mg (per each): \$0.07 - \$0.10

Capsules (Motrin IB Oral)

200 mg (per each): \$0.16

Capsules (Proprinal Oral)

200 mg (per each): \$0.16

Chewable (Advil Junior Strength Oral)

100 mg (per each): \$0.30

Chewable (Motrin Childrens Oral)

100 mg (per each): \$0.29

Kit (Ibupak Oral)

600 mg (per each): \$1,523.00

Solution (Caldolor Intravenous)

800 mg/200 mL (per mL): \$0.16

800 mg/8 mL (per mL): \$4.05

Solution (Ibuprofen Lysine Intravenous)

10 mg/mL (per mL): \$206.25 - \$273.74

Solution (NeoProfen Intravenous)

10 mg/mL (per mL): \$646.22

Suspension (Childrens Advil Oral)

100 mg/5 mL (per mL): \$0.07

Suspension (Childrens Motrin Oral)

100 mg/5 mL (per mL): \$0.06

Suspension (Ibuprofen Oral)

100 mg/5 mL (per mL): \$0.06 - \$0.08

Suspension (Infants Advil Oral)

50 mg/1.25 mL (per mL): \$0.44

Suspension (Motrin Infants Drops Oral)

50 mg/1.25 mL (per mL): \$0.39

Tablets (Aadvaprin Oral)

200 mg (per each): \$0.05

Tablets (Advil Junior Strength Oral)

100 mg (per each): \$0.18

Tablets (Advil Oral)

200 mg (per each): \$0.08

Tablets (IBU Oral)

400 mg (per each): \$0.21

600 mg (per each): \$0.29

800 mg (per each): \$0.38

Tablets (Ibuprofen Oral)

200 mg (per each): \$0.02 - \$0.08

400 mg (per each): \$0.04 - \$3.95

600 mg (per each): \$0.05 - \$4.16

800 mg (per each): \$0.06 - \$4.42

Tablets (Medi-First Ibuprofen Oral)

200 mg (per each): \$0.09

Tablets (Motrin IB Oral)

200 mg (per each): \$0.12

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or

price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Tablet, Oral:

Generic: 600 mg

Administration: Adult

Oral: Administer with food or milk.

Bariatric surgery: Oral solutions may contain nonabsorbable sugars (eg, sorbitol, mannitol, xylitol) that can cause dumping syndrome (Ref). Refer to product labeling.

IV: Calaolor: For IV administration only; infuse over at least 30 minutes (adults).

Preparation for Administration: Adult

Ibuprofen injection (Calaolor): Vials must be diluted prior to use. Dilute with D5W, NS or LR to a final concentration ≤ 4 mg/mL.

Administration: Pediatric

Oral: Administer with food or milk to decrease GI upset.

Oral suspension: Shake suspension well before use. Administer with an accurate measuring device (calibrated oral syringe or measuring cup); do not use a household teaspoon or tablespoon to measure dose (overdosage may occur).

IV:

Ibuprofen injection (Calaolor): For IV administration only.

<18 years of age: Infuse over ≥ 10 minutes.

≥ 18 years of age: Infuse over ≥ 30 minutes.

Ibuprofen lysine injection (NeoProfen): For IV administration only; administration via umbilical arterial line has not been evaluated. Infuse over 15 minutes through IV port closest to insertion site. Avoid extravasation. Do not administer simultaneously via same line with TPN. If needed, interrupt TPN for 15 minutes prior to and after ibuprofen administration, keeping line open with dextrose or saline.

Preparation for Administration: Pediatric

IV:

Ibuprofen injection (Calaolor): Vials must be diluted prior to use. Dilute with D5W, NS, or LR to a final concentration ≤ 4 mg/mL. Available as a premix solution (4 mg/mL) for 800 mg doses.

Ibuprofen lysine injection (NeoProfen): Dilute with dextrose or saline to an appropriate volume. A final concentration of 4 mg/mL or 5 mg/mL has been reported (Ref).

Storage/Stability

Ibuprofen injection (Calaolor): Store intact vials and ready-to-use bags at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Vials must be diluted prior to use. Diluted solutions are stable in D5W, LR, or NS for 24 hours at 20°C to 25°C (68°F to 77°F).

Ibuprofen lysine injection (NeoProfen): Store at 20°C to 25°C (68°F to 77°F); excursions are permitted between 15°C and 30°C (59°F and 86°F). Protect from light. Store vials in carton until use. After first withdrawal from vial, discard remaining solution (preservative free). Following dilution in D5W or NS, use within 30 minutes.

Suspension: Store at 15°C to 30°C (59°F to 86°F).

Tablet: Store at 20°C to 25°C (68°F to 77°F); protect from excessive heat 40°C (104°F).

Use: Labeled Indications

Oral: Management of inflammatory diseases and rheumatoid disorders, mild to moderate pain, fever, dysmenorrhea, and osteoarthritis

Ibuprofen injection (Calabolor): Management of mild to moderate pain, including postoperative pain, and management of moderate to severe pain, including postoperative pain, as an adjunct to opioid analgesics in adults and pediatric patients ≥ 3 months of age; reduction of fever in adults and pediatric patients ≥ 3 months of age.

Ibuprofen lysine injection (NeoProfen): Patent ductus arteriosus: To close a clinically significant patent ductus arteriosus in premature infants weighing between 500 and 1,500 g who are no more than 32 weeks of gestational age when usual medical management (eg, diuretics, fluid restriction, respiratory support) is ineffective.

OTC labeling: Reduction of fever; management of pain due to headache, acute migraine, sore throat, arthritis, or physical or athletic overexertion (eg, sprains/strains), menstrual pain, dental pain, minor muscle/bone/joint pain, backache, and pain due to the common cold and flu.

Use: Off-Label: Adult

Abnormal uterine bleeding, nonacute; Gout, treatment, acute flares; Pericarditis, acute or recurrent

Medication Safety Issues

Sound-alike/look-alike issues:

Motrin may be confused with Neurontin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs (pediatric liquid medications requiring measurement) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Community/Ambulatory Care Settings ©ISMP 2021).

Older Adult: High-Risk Medication:

Beers Criteria: Ibuprofen is identified in the Beers Criteria as a potentially inappropriate medication to be avoided for chronic use in patients 65 years and older (unless alternative agents ineffective and patient can receive concomitant gastroprotective agent) due to increased risk of GI bleeding and peptic ulcer disease in older adults in high-risk category (eg, older than 75 years of age or receiving concomitant oral/parenteral corticosteroids, anticoagulants, or antiplatelet agents). In addition, avoid for short-term scheduled use in combination with oral/parenteral corticosteroids, anticoagulants, or antiplatelet agents unless alternatives are ineffective and patient can receive concomitant gastroprotective agent (Beers Criteria [AGS 2023]).

NSAIDs are identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as potentially inappropriate medications in older adults (≥ 65 years of age) for long-term use in the treatment of gout or osteoarthritis where other agents have not been tried (eg, acetaminophen for osteoarthritis, xanthine-oxidase inhibitors for gout). In addition, some disease states of concern include heart failure, renal impairment, peptic ulcer disease, gastrointestinal bleeding, severe hypertension, and coronary, cerebral or peripheral vascular disease (O'Mahony 2023).

Administration issues:

Injectable formulations: Both ibuprofen and ibuprofen lysine are available for parenteral use. Ibuprofen lysine is **only** indicated for closure of a clinically-significant patent ductus arteriosus.

Metabolism/Transport Effects

Substrate of CYP2C19 (Minor), CYP2C9 (Minor); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential; **Inhibits** OAT1/3

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

5-Aminosalicylic Acid Derivatives: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of 5-Aminosalicylic Acid Derivatives. *Risk C: Monitor*

Abciximab: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Abrocitinib: Agents with Antiplatelet Effects may increase antiplatelet effects of Abrocitinib. *Risk X: Avoid*

Acalabrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Acemetacin: May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Alcohol (Ethyl): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of GI bleeding may be increased with this combination. *Risk C: Monitor*

Aliskiren: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Aliskiren. Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Aliskiren. *Risk C: Monitor*

Aminoglycosides: Nonsteroidal Anti-Inflammatory Agents may decrease excretion of Aminoglycosides. Data only in premature infants. *Risk C: Monitor*

Aminolevulinic Acid (Systemic): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Systemic). *Risk X: Avoid*

Aminolevulinic Acid (Topical): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Topical). *Risk C: Monitor*

Anagrelide: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Angiotensin II Receptor Blockers: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Angiotensin II Receptor Blockers. The combination of these two agents may also significantly decrease glomerular filtration and renal function. Angiotensin II Receptor Blockers may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the combination may result in a significant decrease in renal function. *Risk C: Monitor*

Angiotensin-Converting Enzyme Inhibitors: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Angiotensin-Converting Enzyme Inhibitors. Angiotensin-Converting Enzyme Inhibitors may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the combination may result in a significant decrease in renal function. *Risk C: Monitor*

Anticoagulants (Miscellaneous Agents): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Anticoagulants (Miscellaneous Agents). *Risk C: Monitor*

Antiplatelet Agents (P2Y12 Inhibitors): Agents with Antiplatelet Effects may increase antiplatelet effects of Antiplatelet Agents (P2Y12 Inhibitors). *Risk C: Monitor*

Aspirin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may decrease therapeutic effects of Aspirin. Aspirin may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Specifically, the risk for bleeding may be increased. Aspirin may decrease serum concentrations of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Management: In general, avoid

regular, frequent use of NSAIDs with aspirin whenever possible. If combined, monitor for increased bleeding and a reduced cardioprotective effect of aspirin. *Risk D: Consider Therapy Modification*

Bemiparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Bemiparin. Management: Avoid this combination if possible, due to an increased risk of bleeding. If coadministration cannot be avoided, monitor patients closely for clinical and laboratory evidence of bleeding. *Risk D: Consider Therapy Modification*

Beta-Blockers: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Beta-Blockers. *Risk C: Monitor*

Bile Acid Sequestrants: May decrease absorption of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Bisphosphonate Derivatives: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Bisphosphonate Derivatives. Both an increased risk of gastrointestinal ulceration and an increased risk of nephrotoxicity are of concern. *Risk C: Monitor*

Caplacizumab: Agents with Antiplatelet Effects may increase adverse/toxic effects of Caplacizumab. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Cardiac Glycosides: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Cardiac Glycosides. *Risk C: Monitor*

Clofarabine: OAT1/3 Inhibitors may increase serum concentrations of Clofarabine. *Risk C: Monitor*

Collagenase (Systemic): Agents with Antiplatelet Effects may increase adverse/toxic effects of Collagenase (Systemic). Specifically, the risk of injection site bruising and or bleeding may be increased. *Risk C: Monitor*

Corticosteroids (Systemic): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

CycloSPORINE (Systemic): Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of CycloSPORINE (Systemic). Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of CycloSPORINE (Systemic). CycloSPORINE (Systemic) may increase serum concentrations of Nonsteroidal Anti-Inflammatory Agents. Management: Consider alternatives to nonsteroidal anti-inflammatory agents (NSAIDs). Monitor for evidence of nephrotoxicity, as well as increased serum cyclosporine concentrations and systemic effects (eg, hypertension) during concomitant therapy with NSAIDs. *Risk D: Consider Therapy Modification*

Dasatinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Deferasirox: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Deferasirox. Specifically, the risk for GI ulceration/irritation or GI bleeding may be increased. *Risk C: Monitor*

Deoxycholic Acid: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Desmopressin: Nonsteroidal Anti-Inflammatory Agents may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Dichlorphenamide: OAT1/3 Inhibitors may increase serum concentrations of Dichlorphenamide. *Risk C: Monitor*

Direct Oral Anticoagulants (DOACs): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Direct Oral Anticoagulants (DOACs). *Risk C: Monitor*

Drospirenone-Containing Products: May increase hyperkalemic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Enoxaparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Enoxaparin. Management: Discontinue nonselective NSAIDs prior to initiation of enoxaparin

whenever possible. If coadministration cannot be avoided, monitor patients closely for clinical and laboratory evidence of bleeding. *Risk D: Consider Therapy Modification*

Eplerenone: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Eplerenone. Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Eplerenone. *Risk C: Monitor*

Fluconazole: May increase serum concentrations of Ibuprofen (Systemic). *Risk C: Monitor*

Fondaparinux: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Fondaparinux. Management: Discontinue nonselective nonsteroidal anti-inflammatory agents prior to the initiation of fondaparinux, if possible. If coadministration is required, monitor patients closely for signs and symptoms of bleeding. *Risk D: Consider Therapy Modification*

Glycoprotein IIb/IIIa Inhibitors: Agents with Antiplatelet Effects may increase antiplatelet effects of Glycoprotein IIb/IIIa Inhibitors. *Risk C: Monitor*

Heparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Heparin. *Risk C: Monitor*

Heparins (Low Molecular Weight): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Heparins (Low Molecular Weight). *Risk C: Monitor*

Herbal Products with Anticoagulant/Antiplatelet Effects: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

HydrALAZINE: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of HydrALAZINE. *Risk C: Monitor*

Ibritumomab Tiuxetan: Agents with Antiplatelet Effects may increase antiplatelet effects of Ibritumomab Tiuxetan. *Risk C: Monitor*

Ibrutinib: Agents with Antiplatelet Effects may increase adverse/toxic effects of Ibrutinib. Specifically, the risk of bleeding and hemorrhage may be increased. *Risk C: Monitor*

Imatinib: Ibuprofen (Systemic) may decrease serum concentrations of Imatinib. Specifically, ibuprofen may decrease intracellular concentrations of imatinib, leading to decreased clinical response. Management: Consider using an alternative to ibuprofen in patients who are being treated with imatinib. Available evidence suggests other NSAIDs do not interact in a similar manner. *Risk D: Consider Therapy Modification*

Inotersen: Agents with Antiplatelet Effects may increase adverse/toxic effects of Inotersen. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Ketorolac (Nasal): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Ketorolac (Systemic): Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Ketorolac (Systemic). *Risk X: Avoid*

Limaprost: May increase adverse/toxic effects of Agents with Antiplatelet Effects. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Lithium: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Lithium. *Risk C: Monitor*

Loop Diuretics: Nonsteroidal Anti-Inflammatory Agents may decrease diuretic effects of Loop Diuretics. Loop Diuretics may increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. Management: Monitor for evidence of kidney injury or decreased therapeutic effects of loop diuretics with concurrent use of an NSAID. Consider avoiding concurrent use in CHF or cirrhosis. Concomitant use of bumetanide with indomethacin is not recommended. *Risk D: Consider Therapy Modification*

Lumacaftor and Ivacaftor: May decrease serum concentrations of Ibuprofen (Systemic). *Risk C: Monitor*

Macimorelin: Coadministration of Nonsteroidal Anti-Inflammatory Agents and Macimorelin may alter diagnostic results. *Risk X: Avoid*

MetFORMIN: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of MetFORMIN. *Risk C: Monitor*

Methotrexate: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Methotrexate. Management: Avoid coadministration of higher dose methotrexate (such as that used for the treatment of oncologic conditions) and NSAIDs. Use caution if coadministering lower dose methotrexate and NSAIDs. *Risk D: Consider Therapy Modification*

Methoxsalen (Systemic): Photosensitizing Agents may increase photosensitizing effects of Methoxsalen (Systemic). *Risk C: Monitor*

Methoxyflurane: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Methoxyflurane. *Risk X: Avoid*

Mifamurtide: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Mifamurtide. *Risk X: Avoid*

Miscellaneous Antiplatelets: Agents with Antiplatelet Effects may increase antiplatelet effects of Miscellaneous Antiplatelets. *Risk C: Monitor*

Nadroparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Nadroparin. Management: Coadministration of NSAIDs and nadroparin is not recommended due to an increased risk of bleeding. If coadministration is required, monitor patients closely for clinical and laboratory signs of bleeding. *Risk D: Consider Therapy Modification*

Naftazone: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): Nonsteroidal Anti-Inflammatory Agents (Topical) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of gastrointestinal (GI) toxicity is increased. Management: Coadministration of systemic nonsteroidal anti-inflammatory drugs (NSAIDs) and topical NSAIDs is not recommended. If systemic NSAIDs and topical NSAIDs, ensure the benefits outweigh the risks and monitor for increased NSAID toxicities. *Risk D: Consider Therapy Modification*

Nonsteroidal Anti-Inflammatory Agents: May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk for gastrointestinal toxicity is increased. *Risk X: Avoid*

Obinutuzumab: Agents with Antiplatelet Effects may increase adverse/toxic effects of Obinutuzumab. Specifically, the risk of bleeding may be increased. Management: Consider avoiding coadministration of obinutuzumab and agents with antiplatelet effects, especially during the first cycle of obinutuzumab therapy. *Risk D: Consider Therapy Modification*

Omacetaxine: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Omacetaxine. Specifically, the risk for bleeding-related events may be increased. *Risk C: Monitor*

PEMEtrexed: Ibuprofen (Systemic) may increase serum concentrations of PEMEtrexed. Management: In patients with an estimated creatinine clearance of 45 to 79 mL/min, avoid ibuprofen for 2 days before, the day of, and 2 days following the administration of pemetrexed. Monitor for increased pemetrexed toxicities if combined. *Risk D: Consider Therapy Modification*

Pentosan Polysulfate Sodium: Agents with Antiplatelet Effects may increase adverse/toxic effects of Pentosan Polysulfate Sodium. Specifically, the risk of hemorrhage may be increased. *Risk C: Monitor*

Phenylbutazone: May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Pirtobrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Polyethylene Glycol-Electrolyte Solution: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Polyethylene Glycol-Electrolyte Solution. *Risk C: Monitor*

Porfimer: Photosensitizing Agents may increase photosensitizing effects of Porfimer. *Risk X: Avoid*

Potassium Salts: Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Potassium Salts. *Risk C: Monitor*

Potassium-Sparing Diuretics: Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Potassium-Sparing Diuretics. Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Potassium-Sparing Diuretics. *Risk C: Monitor*

PRALatrexate: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of PRALatrexate. More specifically, NSAIDs may decrease the renal excretion of pralatrexate. Management: Avoid coadministration of pralatrexate with nonsteroidal anti-inflammatory drugs (NSAIDs). If coadministration cannot be avoided, closely monitor for increased pralatrexate serum levels or toxicity. *Risk D: Consider Therapy Modification*

Probenecid: May increase serum concentrations of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Prostaglandins (Ophthalmic): Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Prostaglandins (Ophthalmic). Nonsteroidal Anti-Inflammatory Agents may also enhance the therapeutic effects of Prostaglandins (Ophthalmic). *Risk C: Monitor*

Quinolones: Nonsteroidal Anti-Inflammatory Agents may increase neuroexcitatory and/or seizure-potentiating effects of Quinolones. Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Quinolones. *Risk C: Monitor*

Salicylates: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase adverse/toxic effects of Salicylates. An increased risk of bleeding may be associated with use of this combination. Management: Consider alternatives when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, monitor for gastrointestinal toxicity (eg, gastrointestinal bleeding, ulceration, perforation). *Risk D: Consider Therapy Modification*

Selective Serotonin Reuptake Inhibitor: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Nonsteroidal Anti-Inflammatory Agents (Nonselective) may decrease therapeutic effects of Selective Serotonin Reuptake Inhibitor. Management: Consider alternatives to NSAIDs. Monitor for evidence of bleeding and diminished antidepressant effects. It is unclear whether COX-2-selective NSAIDs reduce risk. *Risk D: Consider Therapy Modification*

Selumetinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Sodium Phosphates: May increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of acute phosphate nephropathy may be enhanced. *Risk C: Monitor*

Sulopenem Etzadroxil: OAT1/3 Inhibitors may increase serum concentrations of Sulopenem Etzadroxil. *Risk C: Monitor*

Sulprostone: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Sulprostone. *Risk X: Avoid*

Tacrolimus (Systemic): Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Tacrolimus (Systemic). *Risk C: Monitor*

Temoporfin: Photosensitizing Agents may increase photosensitizing effects of Temoporfin. *Risk X: Avoid*

Tenofovir Products: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Tenofovir Products. Management: Seek alternatives to these combinations whenever possible. Avoid use

of tenofovir with multiple NSAIDs or any NSAID given at a high dose due to a potential risk of acute renal failure. Diclofenac appears to confer the most risk. *Risk D: Consider Therapy Modification*

Thiazide and Thiazide-Like Diuretics: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Thiazide and Thiazide-Like Diuretics. Thiazide and Thiazide-Like Diuretics may increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Thrombolytic Agents: Agents with Antiplatelet Effects may increase adverse/toxic effects of Thrombolytic Agents. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Tipranavir: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Tolperisone: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Tolperisone. Specifically, the risk of hypersensitivity reactions may be increased. Tolperisone may increase therapeutic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Tricyclic Antidepressants: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents. Tricyclic Antidepressants may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of major adverse cardiac events (MACE), hemorrhagic stroke, ischemic stroke, and heart failure may be increased. *Risk C: Monitor*

Vadadustat: OAT1/3 Inhibitors may increase serum concentrations of Vadadustat. *Risk C: Monitor*

Valproic Acid and Derivatives: Ibuprofen (Systemic) may decrease serum concentrations of Valproic Acid and Derivatives. *Risk C: Monitor*

Vancomycin: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Vancomycin. *Risk C: Monitor*

Verteporfin: Photosensitizing Agents may increase photosensitizing effects of Verteporfin. *Risk C: Monitor*

Vitamin E (Systemic): May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Vitamin K Antagonists: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Volanesorsen: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Voriconazole: May increase serum concentrations of Ibuprofen (Systemic). Specifically, concentrations of the S-(+)-ibuprofen enantiomer may be increased. *Risk C: Monitor*

Zanubrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Reproductive Considerations

Nonsteroidal anti-inflammatory drugs (NSAIDs) may delay or prevent rupture of ovarian follicles. This may be associated with infertility that is reversible upon discontinuation of the medication. Consider discontinuing use in patients having difficulty conceiving or those undergoing investigation of fertility.

Based on available information, NSAIDs can be continued in patients with rheumatic and musculoskeletal diseases who are planning to father a child (ACR [Sammartano 2020]).

Pregnancy Considerations

The use of nonsteroidal anti-inflammatory drugs (NSAIDs) close to conception may be associated with an increased risk of miscarriage due to cyclooxygenase-2 inhibition interfering with implantation (Bermas 2014; Bloor 2013).

Birth defects have been observed following in utero NSAID exposure in some studies; however, data are conflicting (Bloor 2013). Nonteratogenic effects, including prenatal constriction of the ductus arteriosus, persistent pulmonary hypertension of the newborn, oligohydramnios, necrotizing enterocolitis, renal dysfunction or failure, and intracranial hemorrhage, have been observed in the fetus/neonate following in utero NSAID exposure (Bermas 2014; Bloor 2013). Maternal NSAID use may cause fetal renal dysfunction leading to oligohydramnios. Although rare, this may occur as early as 20

weeks' gestation and is more likely to occur with prolonged maternal use. Oligohydramnios may be reversible following discontinuation of the NSAID (Dathe 2019; FDA 2020). In addition, nonclosure of the ductus arteriosus postnatally may occur and be resistant to medical management (Bermas 2014; Bloor 2013).

Avoid maternal use of NSAIDs beginning at 20 weeks' gestation. If NSAID use is necessary between 20 and 30 weeks' gestation, limit use to the lowest effective dose and shortest duration possible; consider ultrasound monitoring of amniotic fluid if treatment extends beyond 48 hours, and discontinue the NSAID if oligohydramnios is found (FDA 2020). Because NSAIDs may cause premature closure of the ductus arteriosus, prescribing information for ibuprofen specifically states to avoid use starting at 30 weeks' gestation.

Based on available information, NSAIDs can be continued during the first 2 trimesters of pregnancy in patients with rheumatic and musculoskeletal diseases; use in the third trimester is not recommended (ACR [Sammaritano 2020]).

Treatment for migraine headaches in pregnant patients should be individualized (AHS [Ailani 2021]). The acute treatment of migraine headaches during pregnancy should be initiated with an agent other than an NSAID. If an NSAID is needed as second-line therapy, treatment should be limited to the second trimester and total duration of therapy should be no longer than 48 hours. Ibuprofen may be preferred when an NSAID is needed (ACOG 2022).

NSAIDs may be used as part of a multimodal approach to pain relief following cesarean delivery (ACOG 2019).

Breastfeeding Considerations

Ibuprofen is present in breast milk.

Data related to the presence of ibuprofen in breast milk are available from multiple studies (Rigourd 2014; Townsend 1984; Walter 1997; Weibert 1982).

- According to the manufacturer, the relative infant dose (RID) of ibuprofen is 0.06% to 0.6% based on the weight-adjusted maternal dose.
- Six doses of oral ibuprofen 400 mg were administered to a lactating patient for pain following maxillary surgery (postpartum age not stated). Thirty minutes following administration of the first dose, breast milk concentrations were 13 ng/mL. The highest concentration of ibuprofen in breast milk was 181 ng/mL measured after 20.5 hours. Using the actual weight of the mother and infant, the authors calculated the RID of ibuprofen to be 0.0008% (Walter 1997).
- Ibuprofen concentrations in breast milk are available from 13 lactating mothers (mean GA at delivery: 36.6 weeks) with a duration of breastfeeding between 12 and 1,270 days. Ibuprofen doses ranged from 400 mg to 1,200 mg/day (mean dose: 1,046 mg/day). The mean concentration of ibuprofen in breast milk was 361.23 mcg/L (range 163.95 to 590.08 mcg/L). Authors of the study calculated the estimated infant dose of ibuprofen via breast milk to be 68 mcg/kg/day (range: 8 to 262 mcg/kg/day) providing a RID <0.38% of the weight-adjusted maternal dose (range 0.04% to 1.53%). Protein content of breast milk decreases over time and ibuprofen is highly protein bound. It was observed in this study that the transfer of ibuprofen into milk decreased with the protein content as well as the duration of lactation. The highest relative infant doses were calculated in premature infants (1.136% and 1.531%) who were both <30 days old during the study (Rigourd 2014).
- In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

A prospective cohort study evaluated the outcomes of breastfed infants whose mothers were taking various medications. Within the study, 21 mother-infant pairs reported ibuprofen exposure (dose, duration, relationship to breastfeeding not provided). There were no cases of diarrhea, drowsiness, or irritability in the breastfed infants (Ito 1993). Based on the available data, adverse events have not been reported in breastfeeding infants and milk production is not affected.

Ibuprofen is considered compatible with breastfeeding when used in usual recommended doses (WHO 2002). Nonopioid analgesics, including nonsteroidal anti-inflammatory drugs (NSAIDs), are preferred

for breastfeeding patients who require pain control peripartum or for surgery outside of the postpartum period (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]). NSAIDs are considered compatible for the treatment of rheumatic and musculoskeletal diseases in lactating patients; ibuprofen is the preferred NSAID (ACR [Sammaritano 2020]). NSAIDs may be used to treat acute migraine in lactating patients (ACOG 2022).

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother. Avoid maternal use of NSAIDs if the breastfeeding infant has platelet dysfunction, thrombocytopenia, or a ductal-dependent cardiac lesion (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; Bloor 2013).

Dietary Considerations

Some products may contain phenylalanine and/or potassium.

Monitoring Parameters

CBC, chemistry profile, occult blood loss and periodic LFTs; monitor response (pain, range of motion, grip strength, mobility, ADL function), inflammation; observe for weight gain, edema; monitor renal function (urine output, serum BUN and creatinine); observe for bleeding, bruising (especially in patients with coagulation disorders or who are receiving anticoagulants); monitor for anemia with long-term therapy; evaluate GI effects (abdominal pain, bleeding, dyspepsia); mental confusion, disorientation; BP; periodic ophthalmic exams with long-term therapy; signs of infection (ibuprofen lysine); signs of immediate or delayed hypersensitivity reactions.

Reference Range

Plasma concentrations >200 mcg/mL (SI: >970 micromole/L) may be associated with severe toxicity.

Mechanism of Action

Reversibly inhibits cyclooxygenase-1 and 2 (COX-1 and 2) enzymes, which results in decreased formation of prostaglandin precursors; has antipyretic, analgesic, and anti-inflammatory properties

Other proposed mechanisms not fully elucidated (and possibly contributing to the anti-inflammatory effect to varying degrees), include inhibiting chemotaxis, altering lymphocyte activity, inhibiting neutrophil aggregation/activation, and decreasing proinflammatory cytokine levels.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Oral: Analgesic: Within 30 to 60 minutes (Davies 1998; Mehlisch 2013); Antipyretic: Single oral dose 8 mg/kg (Kauffman 1992): Infants ≤ 1 year: 69 ± 22 minutes; Children ≥ 6 years: Single oral dose 8 mg/kg (Kauffman 1992): 109 ± 64 minutes; Adults: <1 hour (Sullivan 2011).

Maximum effect: Antipyretic: 2 to 4 hours.

Duration: Oral: Antipyretic: 6 to 8 hours (Sullivan 2011).

Absorption: Oral: Rapid (85%).

Distribution: V_d :

Oral: Febrile children <11 years: 0.2 L/kg; Adults: 0.12 L/kg.

IV: Ibuprofen (Caldolor):

Pediatric patients 6 months to <2 years: 0.31 L/kg.

Pediatric patients 2 to 16 years: 0.23 L/kg.

IV: Ibuprofen lysine: Premature neonates, GA <32 weeks: Variable results observed: 0.32 L/kg, others have reported: a central compartment V_d that decreases with increasing PNA and ductal closure (Van Overmeire 2001) and a V_d , apparent: 0.062 L/kg in 21 premature neonates (GA <32 weeks, PNA: <1 day) (Aranda 1997).

Protein binding: >99%; Premature infants: ~95% (Aranda 1997).

Bioavailability: 80%.

Metabolism: Hepatic via oxidation; **Note:** Ibuprofen is a racemic mixture of R and S isomers; the R isomer (thought to be inactive) is slowly and incompletely (~60%) converted to the S isomer (active) in adults; the amount of conversion in children is not known, but it is thought to be similar to adults; a study in preterm neonates estimated the conversion to be 61% after prophylactic ibuprofen use and 86% after curative treatment (Gregoire 2004).

Half-life elimination: IV:

Ibuprofen (Caldor):

Pediatric patients:

Infants 3 months to <6 months: 1.3 hours.

Infants ≥6 months to Children <2 years: 1.8 hours.

Children ≥2 years to Adolescents ≤16 years: ~1.5 hours.

Adults: 2.22 to 2.44 hours.

Ibuprofen lysine (Neoprofen):

Premature neonates, GA <32 weeks: Reported data highly variable.

R-enantiomer: 10 hours; S-enantiomer: 25.5 hours (Gregoire 2004).

Age-based observations:

PNA <1 day: 30.5 ± 4.2 hours (Aranda 1997).

PNA 3 days: 43.1 ± 26.1 hours (Van Overmeire 2001).

PNA 5 days: 26.8 ± 23.6 hours (Van Overmeire 2001).

Half-life elimination: Oral:

Children 3 months to 10 years: Oral suspension: 1.6 ± 0.7 hours (Kauffman 1992).

Adults: ~2 hours; End-stage renal disease: Unchanged (Aronoff 2007).

Time to peak: Tablets: 1 to 2 hours; Suspension: 1 hour.

Children with cystic fibrosis (Scott 1999):

Suspension (n=22): 0.74 ± 0.43 hours (median: 30 minutes).

Chewable tablet (n=4): 1.5 ± 0.58 hours (median: 1.5 hours).

Tablet (n=12): 1.33 ± 0.95 hours (median: 1 hour).

Excretion: Urine (primarily as metabolites (45% to 80%); ~1% as unchanged drug and 14% as conjugated); some feces.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and at NSAIDs Class Medication Guide, must be dispensed with this medication.

Patient Counseling Points

(For additional information see "Ibuprofen: Patient drug information")

What is this drug used for?

- It is used to ease pain, swelling, and fever.
- It is used to ease painful period (menstrual) cycles.
- It is used to treat arthritis.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Constipation, diarrhea, stomach pain, upset stomach, or throwing up
- Heartburn
- Gas
- Dizziness

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Aseptic meningitis like headache, fever, chills, very upset stomach or throwing up, stiff neck, rash, bright lights bother your eyes, feeling sleepy, or feeling confused
- Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes
- Bleeding like throwing up or coughing up blood; vomit that looks like coffee grounds; blood in the urine; black, red, or tarry stools; bleeding from the gums; abnormal vaginal bleeding; bruises without a cause or that get bigger; or bleeding you cannot stop
- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath
- High blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Shortness of breath, a big weight gain, or swelling in the arms or legs
- Chest pain or pressure or a fast heartbeat
- Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight
- Feeling very tired or weak
- Ringing in ears
- Very bad back pain
- Change in eyesight
- Severe skin reactions; red, swollen, blistered, or peeling skin; red or irritated eyes; sores in the mouth, throat, nose, eyes, genitals, or any areas of skin; fever; chills; body aches; shortness of breath; or swollen glands
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT

include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Brufen | Dolgit | Doloraz | Emifen | Fenbid | I profen | Ibucare | Ibugesic | Ibulgan | Ibusoft | Irfen | Jazofen | Medafen | Monoflam | Neoprofen | Nufen | Prof | Profinal | Sapofen | Spedifen;
- (AR) Argentina: Actron | Actron 600 rapida accion | Actron mujer forte rapida accion | Actron pediatrico | Actron rapida accion | Acuifem | Acuifem 15 | Advil | Afebril | Afebril masticable | Algioprofen | Atomo desinflamante ibu | Baydol | Baydol mujer rapida accion | Bistryl | Buidiona | Causalon ibu | Copiron | Dolorsyn termico | Drusel vl | Fabogesic | Fabogesic ninos | Fabogesic rapida accion | Fabogesic vl | Fada ibuprofeno | Febratic | Fiebrovl | Fontol | Ibu | Ibu apracur | Ibu apracur forte | Ibu multin | Ibubenitol | Ibucler | Ibudolor | Ibuevanol forte rapida accion | Ibuevanol rapida accion | Ibufabra | Ibufec | Ibufix | Ibufull | Ibukasec | Ibulam | Ibulgia | Ibumar | Ibumax | Ibumejoral | Ibumejoral ninos | Ibumultin | Ibup | Ibupirac | Ibupirac pediatrico | Ibupiretas | Ibuprofenix mujer | Ibuprofeno | Ibuprofeno biocrom | Ibuprofeno bouzen peneco | Ibuprofeno elisium | Ibuprofeno elisium mujer | Ibuprofeno fecofar | Ibuprofeno fecofar vl | Ibuprofeno ilab | Ibuprofeno iraola | Ibuprofeno klonal | Ibuprofeno klonal forte | Ibuprofeno klonal forte nf | Ibuprofeno klonal max | Ibuprofeno klonal nf | Ibuprofeno lafedar | Ibuprofeno larjan | Ibuprofeno puntanos | Ibuprofeno purissimus | Ibuprofeno richet | Ibuprofeno richet vl | Ibuprofeno rospaw | Ibuprofeno rospaw vl | Ibuprofeno sant gall friburg | Ibuprofeno tauro | Ibuprofeno tauro vl | Ibuprofeno trb pharma | Ibuprofeno valmax | Ibuprofeno vent 3 | Ibusi | Ibusol | Ibusol anana | Ibusol forte | Ibusol pediatrico | Ibusumal | Ibutafiol | Ibutenk | Ibutenk forte | Ibutenk vl | Ibuxfar | Ibuxim | Ibuzidine | Labsyna | Matrix nucleo disgregable ra | Mejoral ibuprofeno rapida accion | Motrax | Novogeniol | Novogeniol rapida accion | Pakurat | Ponstin | Ponstinetas | Si end | Sindol | Sindol 400 vl | Termokids | Tonal;
- (AT) Austria: Adolorin ibuforte direkt | Adolorini direkt | Aktren | Kratalgin | Neobrufen rapid | Nureflex | Nurofen | Pedeia | Ratiodolor rapid | Thomaprodol;
- (AU) Australia: Adiramedica ibuprofen | Advil | Advil 400 double strength | Advil children 2-12 years pain & fever relief | Advil children 2-7 years pain & fever relief | Advil children 7-12 years pain & fever relief | Advil childrens 7-12 years | Advil infant 3 months-2 years pain & fever relief | Advil infants pain & fever relief | Amcal ibuprofen | Apohealth children's ibuprofen | Apohealth children's ibuprofen infant | Apohealth childrens ibuprofen | Apohealth ibuprofen | Blooms the chemist ibuprofen | Brufen | Bugesic | Chemists own ibuprofen | Chemists' own children's ibuprofen infant | Chemists' own children's ibuprofen pain & fever 6 months - 12 years | Chemists' own children's ibuprofen pain & fever suspension 6 months - 12 years | Chemists' own ibuprofen | Chemists' own ibuprofen double strength | Chemists' own ibuprofen long lasting | Chemists' own ibuvance | Chemmart pharmacy ibuprofen | Children's ibuprofen | Cipla pain relief for little kids ibuprofen | Coles ibuprofen | Discount drug stores ibuprofen | Ethics ibuprofen | Fenpaed | Fenpaed double strength | Gofex | Hedafen ibuprofen | Help@hand for pain & inflammation ibuprofen | Herron blue | Herron blue kids 7+ | Herron ibuprofen blue | I ibuprofen | Ibusoft | IbuVance | Medichoice ibuprofen | Medichoice ibuprofen for kids | Medichoice ibuprofen minicaps pain and inflammation | Medihealth ibuprofen | Medix ibuprofen | Medix ibuprofen double strength | Medix ibuprofen mini caps pain & inflammation | Mendeleev children's ibuprofen | Neo health ibuprofen | Neofen red | Neohealth ibuprofen | Noumed ibuprofen suspension for children 3 months to 5 years | Noumed ibuprofen suspension for children 5 to 12 years | Nurofen for children | Nurofen for children 3 months - 5 years | Nurofen for children 5-12 years |

- Nurofen for children 7+ years | Nurofen for children baby 3+ months | Nurofen junior | Nurofen zavance | Nyal ibuprofen | Panafen ib | Pc ibuprofen | Pharmacy care ibuprofen | Pharmacy choice ibuprofen | Pharmacy choice ibuprofen children's | Pharmacy health ibuprofen | Pharmacy health ibuprofen suspension for baby 3 months+ | Pharmacy health ibuprofen suspension for children 3 months to 5 years | Pharmacy health ibuprofen suspension for children 5 to 12 years | Pharmacy own ibuprofen | Priceline ibuprofen | Priceline pharmacy ibuprofen | Priceline pharmacy ibuprofen suspension for children 3 months to 5 years | Priceline pharmacy ibuprofen suspension for children 5 to 12 years | Proven | Rafen | Soul pattinson ibuprofen | Spiro lifecare ibuprofen | Terry white chemists ibuprofen | Terry white chemmart ibuprofen | Terrywhite chemmart ibuprofen | Terrywhite chemmart ibuprofen suspension for children 3 months to 5 years | Terrywhite chemmart ibuprofen suspension for children 5 to 12 years | Trust for kids ibuprofen infant drops | Trust ibufen | Trust ibuprofen | Wagner health children's ibuprofen | Wagner Health Childrens Ibuprofen | Wagner Health Ibuprofen;
- (BD) Bangladesh: Advel | Antiflam | Arafa | Bufen | Chemofen | Deprofen | Diprofen | Erofen | Esrufen | Flam | Flamex | Flamic | Ibufen | Ibuflam | Iburex | Ibutab | Inflam | Neoflam | Noflam | Norfen | Profen | Rapofen | Rebufen | Reufen | Reumafen | Serviprofen | Tyflam;
 - (BE) Belgium: Algidrin Pediatrico | Brufen | Bufedon | Ibu slow | Ibucaps | Ibuprofen eg | Ibuprofen ratiopharm | Ibuprofen sandoz | Ibuprofen teva | Junifen | Malafene | Motrin | Nurofen | Perdotensio;
 - (BF) Burkina Faso: Bufalgan | Ibufen | Ibufil | Ibuprofene | Ibuprofene tm | Ibuprofene ubigen | Iprofene | Lexofen | Paidofebril | Trifene;
 - (BG) Bulgaria: Blokmax for kids | Brufen | Flexistad junior | Ibudolor | Ibufen | Ibuprom | Ibuprom optima | Ibuprom sprint | Ibuprom sprint max | Iburapid kids | Mig | Mig junior | Milofen kids | Nurofen | Nurofen express | Nurofen kids | Panactiv | Panactiv forte | Parabol max | Pedeia | Syafen | Upfen;
 - (BR) Brazil: Adlyv | Advil | Algi reumatril | Algy flanderil caps | Algy gotas | Algy-flanderil | Alivium | Aludor | Alyviazes | Artil | Atrofem | Buprolivium | Buprovil | Buscofem | Buscofem girls | Capsfen | Dalsy | Doraliv | Febsen | Ibucaps | Ibuflex | Ibufran | Ibuglobo | Ibuliv | Ibupril | Ibuprofeno | Ibupromed | Ibuprotrat | Ibuprotrat cm | Ibuprotrat infantil | Ibufix | Lombalgina | Medmulher | Motrin | Novalfem | Otiun | Pralivio | Pralivio mulher | Sanafen | Spidufen | Termomed ibup | Vantil;
 - (BZ) Belize: Advil liqui gels | Ibudel | Ibuprofeno;
 - (CH) Switzerland: Afelol liquid caps | Algifor | Algifor dolo forte | Algifor Dolo Junior | Brufen | Iproben | Irfen | Irfen dolo | Optifen | Optifen Junior | Spedifen;
 - (CI) Côte d'Ivoire: Aavil | Algofene | Bufalgan | Bufenol | Dolven | Ibu care | Ibuprofene | Ibusoft | Iprofene | Paidofebril | Pedifen;
 - (CL) Chile: Actron | Actron pediatrico | Actronra | Adfen | Alginasol forte | Bediatil | Bediatil forte | Bladex | Deucodol | Dolad | Dolgex | Esanterm | Ibu | Ibu 2 | Ibucalm | Ibufas | Ibuflam | Ibuflam forte | Ibufren | Ibukids forte | Ibupirac | Ibuprofeno | Ipson | Niofen | Niofen forte | Painpac | Pediaprofen | Pironal | Pyriped | Yetro;
 - (CN) China: Abrofen | An rui ke | BANG QI | Bo fei te | Di er luo | Fen ke | Fen ni kang | Ji hao | Jian de lin | Jin lai fen bu de | Motrin | Si bai de | Spedifen | Tai bao | Tuo en | Ya wei;
 - (CO) Colombia: Actron | Actron max | Advil children | Advil fastgel | Advil max | Aavil teens | Alcadol | Alkadol | Asulpin | Biopen | Biprofen | Bufedol | Buprofen | Buri | Dafenal | Diantal | Diantal pediatrico | Dol | Dolgud | Doliflam | Dolinstant | Dolivium | Dolivium flash | Dolonet | Dolorsin | Dolotrin children | Doraliv | Espidifen | Fai dol | Febrifen | Femen | Fenpic | Freedol | Frenidol | Gelpiron | Gofen forte | H-Ioniten | Ibudol | Ibudol soft | Ibuflash | Ibuflash forte | Ibumer | Ibupiretas | Ibupirino | Ibuprofar | Ibuprofeno | Ibuprofeno bcn | Ibuprofeno mk | Infibu | Infredos | Kalmadol children | Kalmadol elc | Ketono | Lutifen | Mejoral ultra | Mejoralito | Mejorfex | Motrin | Multidol | Multidol express | Niofen | Notts doraliv ninos | Oren | Pedeia | Pedialab fever fast | Pire fen | Pirefen | Profex | Promofen | Qbedol | Rodalgin | Smadol | Suprifeno | Trosifen | Vencedol | Vonprofen;
 - (CR) Costa Rica: Dorival | Febrilex | Febrin | Ibuprofeno | Ibuprofeno mk | Ibuwin | Iprofene | Pedeia | Pyriped;
 - (CZ) Czech Republic: Aavil | Apo ibuprofen | Brufen | Ibalgin | Iboval rapid | Ibuberl | Ibudolor | Ibuprofen apotex | Ibuprofen dh | Ibuprofen dr.max | Ibuprofen farmalider | Inflanor | Iprofenex

- | Nurofen | Nurofen pro deti | Nurofen pro deti jahoda | Pedea | Solpaflex;
- (DE) Germany: Aktren | Aktren spezial | Anco | Contraneural | Dignoflex | Dolgit | Dolo puren | Dolormin | Dolormin instant | Dysdolen | Esprenit | Eudorlin ibuprofen | Exneural | Ibu | Ibu 1a pharm | Ibu benuron | Ibu kd | Ibubeta | Ibuflam | Ibufug | Ibuhexal | Ibuphlogont | Ibuprofen berco | Ibuprofen ct | Ibuprofen milinda | Ibuprofen sandoz | Ibuprofen stada | Ibuprofen temmler | Ibutad | Ibutad akut | Ilvico grippal | Imbun | Jenaprofen | Lesss | Medokal | Nurofen | Optalidon ibu | Parsal | Pedea | Proff schmerzkapseln | Pyracophen | Schmerz dolgit | Seclodin | Spalt liqua | Spalt mobil;
- (DO) Dominican Republic: Aavil max | Artran | Colbufen | Espidifen | Femmex plus | Frevac | Iberofeno | Ibucedin | Ibufar | Ibufen | Ibukid | Ibumen | Ibuprofen feltrex | Ibuprofen Inmenol | Ibuprofeno | Ibuprofeno exels | Ibuprofeno mk | Ibuwin | Ibuwin forte | Iferin | Inafeb | Suprofen teen;
- (EC) Ecuador: Adwin | Adwin forte | Afebril | Analgion | Analgion forte | Analgion infantil | Bediatil | Buprex | Buprex flash | Buprex forte | Disfebral | Disfebral fast | Dolalgial | Dolcaps | Dolfemin | Dolochek plus | Dolonet | Dolonet forte | Dolor out | Dolorgesic | Dolorgesic forte | Dolowin | Dolowin forte | Dolowyn | Doloxen | Efficol forte | Espidifen | Fabogesic rapida accion | Febertt forte | Febroxial | Femen | Femen forte | Gemprofen forte | Hg iprofen | Ibu letic | Ibucalm | Ibufen | Ibuflash | Ibuflash forte | Ibugen | Ibugesic | Ibuprofeno | Ibuprofeno ariston | Ibuprofeno genfar | Ibuprofeno mk | Ibuprofeno nifa | Ibuprofeno ninos mk | Ibuprol | Ibupront | Ibupront forte | Ibuprofotis | Ibupstar | Ibusol | Ibutron | Ibutron flash | Ibuwin | Ibuwin forte | Ibuwin plus | Iprofen | Ipson | Itugin | Krisfhen | Krishfen | Kroxifeno | Marbux | Motrin | Multidol | Nefor | Probinex | Profem | Profimtep | Profinal | Profinal flash | Profinal forte | Pyriped | Pyriped forte | Rafican | Raserdol | Sindlor | Sine dol | Sinfebril | Sinfebril forte | Tasifen | Tenvalin | Tenvalin forte | Termyl forte;
- (EE) Estonia: Advil | Brufen | Brumare | Forsium | Ibufen | Ibuhexal | Ibumax | Ibumetin | Ibusan | Ibustar | Ipren | Larofen lastele | Motrin | Neoprofen | Noritis | Nurofen | Nurofen strawberry | Nypren | Solpaflex;
- (EG) Egypt: Alphafen | Anabruf | Brufen | Flabu | Flamotal | Ibucalmin | Ibufen | Ibuhexal | Juspoled | Mafo | Marcofen | Maxiprofen | Megafen n | Peopobruf | Pradol | Profinal | Profusol | Rapifen | Ultrafen;
- (ES) Spain: Alginasdin | Algidrin infantil | Algidrin Pediatrico | Apirofeno | Asmara | Babypiril | Dalsy | Dalsydol | Dolorac pediatrico | Doltra | Escarmine | Espididol | Fiedosin | Gelofeno | Geloprofen | Ibubex | Ibudol | Ibudol pediatrico | Ibufarmalid | Ibufen infantil | Ibukern | Ibumac | Ibuprofeno aldo union | Ibuprofeno apotex | Ibuprofeno benel | Ibuprofeno cinfa | Ibuprofeno codramol | Ibuprofeno davur | Ibuprofeno durban | Ibuprofeno farmasierra | Ibuprofeno kern | Ibuprofeno kern pharma | Ibuprofeno Normon | Ibuprofeno ratiopharm | Ibuprofeno sandoz | Ibuprofeno stadapharm | Ibuprofeno tarbis farma | Ibuprofeno teva | Ibuprofeno winadol | Junifen | Neobrufen | Neobrufen retard | Nodolfen | Nurofen | Nurofen rapid | Paidofebril | Pedea | Ratiadol;
- (ET) Ethiopia: Betafen | Gofen | Profen;
- (FI) Finland: Brufen | Burana | Femapirin | Ibuxin | Ibuxin caps;
- (FR) France: Advilcaps | Advilmed | Antarene | Gelufene | Ibupradoll | Ibuprofene biogaran | Ibuprofene mylan | Ibuprofene mylan conseil | Ibuprofene RPG | Nureflex | Nurofen | Nurofencaps | Nurofenpro | Spifen;
- (GB) United Kingdom: Aavil liqui gels | Anadin ibuprofen | Anadin Joint Pain | Anadin liquifast | Anadin ultra | Boots ibuprofen 3 months plus | Boots migraine pain relief | Brufen | Calprofen | Cuprofen | Dexcel Ibuprofen | Femafen | Feminax Express | Fenbid | Fenpaed | Feverfen | Flarin | Flarin Joint & Muscular Pain Relief | Fleximex | Galpharm ibuprofen | Galpharm migraine relief | Galprofen | Ibrufhalal | Ibu slo | Ibucalm | Ibular | Ibuprofen kent | Ibuprofen sandoz | Inabrin | Mandafen | Maxagesic | Novaprin | Novaprin nova hc | Nurofen | Nurofen (GSL) | Nurofen (P) | Nurofen back pain | Nurofen express | Nurofen for children strawberry | Nurofen for Children Strawberry Singles | Nurofen joint & back pain relief | Nurofen Maximum Strength Migraine Pain | Nurofen migraine | Nurofenpro | Orbifen | Proflex | Radian b | Seclodin | Suspren | T&R ibuprofen | Teva migraine relief;
- (GH) Ghana: Anfen | ENAFEN | Letafen;
- (GR) Greece: Advil | Algofren | Buscofem | Nurofen for children;
- (GT) Guatemala: Agefen | Agefen fem | Agefen forte | Agefen kids | Asmara | Baykid dolor y

- fiebre | Bifen | Dorival | Gelprofen | Ibuflash forte | Ibun | Ibuprofeno | Ibutanz | Ibutwin | Ibutwin forte | Ibutwin mujer | Nanprofen;
- (HK) Hong Kong: Aavil | Bifen | Brufen | Dae hwa ibuprofen | Febryn | Ibrufen | Ibufac | Infacalm | Jecefarma ibuprofen | Korus ibuprofen | Neutropain | Nurofen | P fen | Parkins | Perofen | Potofen | Profen | Schufen | Slow ibuprofen | Tai guk ibuprofen | Tospan ibuprofen | Trifene | Uni ibuprofen | Vicbrofen | Willipo ibuprofen;
 - (HR) Croatia: Brufen | Brufen Effect | Dalsy | Dalsy forte | Fidiprofen | Ibuprofen alkaloid | Ibuprofen belupo | Ibuprofen farmal | Ibuprofen Pliva Forte | Ibustar | Neodol | Neodol forte | Neofen | Nurofen | Nurofen forte | Nurofen rapid forte | Nurofen za djecu | Peda | Rapidol | Rapidol S | Spidifen;
 - (HU) Hungary: Aavil ultra | Algoflex | Algoflex rapid | Algoflex rapid mini | Brufen | Dolgit | Dolgit akut | Dolowill baby | Ibuthexal | Ibulos | Ibuprofen aktiv | Ibuprofen boehringer ingelheim | Ibustar | Novorufen | Nurofen | Peda | Rapidophen | Solpaflex;
 - (ID) Indonesia: Anafen | Arfen | Arthrifin | Axofen | Brufen | Bufect | Bunofa | Dofen | Dolofen-f | Ethifen | Farsifen | Febryn | Fenatic | Fenbid | Fenris | Hufagripp tmp | Ibufen | Ifen | Iprox | Lexaprofen | Mofen | Moris | Motrin | Nofena | Novaxifen | Ostarin | Prifen | Profen | Proris | Prosic | Prosinal | Provinas | Rhelafen | Ribunal | Shelofen | Stelar | Xepafen forte | Yariven | Zentarin;
 - (IE) Ireland: Brufen | Brufen paediatric | Brupro | Bufigen | Buplex junior | Easofen for children strawberry | Fenopine | Nurofen for children cold & flu & pain orange | Nurofen for children cold & flu & pain strawberry | Nurofen rapid relief | Provin;
 - (IL) Israel: Adex | Ibufen;
 - (IN) India: Alfam | Arfen | Bren | Brufen | Brunex | Cipgesic | Fenlong | Ibutwell | Ibutid | Ibuton | Ibutamar | Ibutgesic | Ibutar | Ibutspan | Ibutsynth | Ibutas | Myofen | Neurophen | Reducin | Tricoff | Tudofen;
 - (IQ) Iraq: Brufemel | Ibuprofen awa | Ibuprofen mdi | Kinprofen | Profedain | Profedin | Profen | Rexofin;
 - (IS) Iceland: Alvofen express | Ibufen;
 - (IT) Italy: Algidrin | Algopirina febbre e dolore | Antalfort | Arfen | Arfen pronto | BENATIA Bambini | Brufen | Brufenlik | Buscofen | BuscofenAct | Calmine | Dolibloc | Doliprokid | Dolofast | Edenil | Fenkid | Fevral | Fluibron febbre e dolore | Fluifort febbre e dolore bambini | Ganapofene | Gineflor | Ginenorm | Ibutbaby | Ibutlev bambini | Ibuprofene aurobindo pharma | Ibuprofene doc generici | Ibuprofene dr. max | Ibuprofene eg | Ibuprofene sandoz | Ibuprofene zentiva | Kendo | Levifen | Moment | Momentact | Momentact analgesico | Momentkid | Nurofast | Nurofen febbre e dolore | Nurofen febre e dolore | Nurofencaps | Pavik | Peda | Rofixdol febbre e dolore | Sinifev | Spididol Analgesic | Spididolpocket | Vicks febbre e dolore;
 - (JO) Jordan: Balkaprofen | Brufen | Doloraz | Fenbid | Fleximex | Ibutgesic | Ibutphil | Ibutram | Isofen | Jofen | Midofen | Nashat | Profast | Profast forte | Remofen max | Rupan | Sapofen | Taskine;
 - (JP) Japan: Anflagen | Bluton | Bruf | Brufen | Buburone | Butylenin | Chilka chugai | Fera | Fescholin | I.b. | Ibutufen | Ibutramen | Ibutief | Ibutprocin | Ibuprofen amel | Ibuprofen mita | Ibuprofen ohara | Ibuprofen sawai | Ibuprofen tatum | Ibuprofen tsuruhara | Iputes | Lamidon kowa | Landelun | Landelun tsuruhara | Liptan merck hoei | Liptan taiyo | Meridon ib | Mynosedin | Nagifen isei | Nagifen s.s. seiyaku | Napacetin | New kerorin e | Nobfen | Nobgen | Nobgen ace | Ringl ib | Sednafen taisho | Unipron;
 - (KE) Kenya: Asfen fc | Betafen | Bren | Brufen | Brufen paediatric | Brufen retard | Brufol | Brumed | Brunex | Calfen | Daprofen | Fenpro | Gofen | Ibu | Ibucos | Ibutfast | Ibutfen | Ibutfil | Ibutflam | Ibutlite | Ibutmex | Ibutn | Ibutrin | Ibut | Ifen | Markvil | Medifen | MI ibuprofen | Neoprofane | Nurofen | Nurofen migraine | Okagesic | Orbifen for children | Ossfen | Pedifen | Ponafen | Profen | Triofen | Ubimol | Zorafen;
 - (KR) Korea, Republic of: Advil | Albofen bill | Allopan | Allyfen | Alopam | Alupen | Arfen | Bebyzal | BR Ibutfen | Brepin s | Brongfen | Brufen | Budipan | Burofon | Carol | Carol f | Champ ibufen | Cleanfen | Cobo | Coldaewon kids ibufen | Coldaewon zero ibufen | Comet | Coritussal f | Dafen | Dafen q | Dexfen | Diapen | Dongkoo ibuprofen | Dosifen | E q pen kids i | Encafen | Encapen | Eve one q | Exipren | Ezn 6 any | Fullopain | Greenfen | Haispen | Hanopen | I profen f | Ibu | Ibu oneq s | Ibu x | Ibucen | Ibucoldben | Ibutuo | Ibutfen | Ibutfenon | Ibunal | Ibunewfen | Ibutone qs | Ibuprofen cho a | Ibuprofen dongindang | Ibuprofen dongyang | Ibuprofen hana | Ibuprofen hangook | Ibuprofen samsung | Ibuprofen samwoo | Ibutoen | Ibutofen | Ibuton | Ibutons | Ibutus

- | Ibususpen | Ifenosin | Inberol | Inbromin | Ipreen | Irofein | Irofen | Ivupan | Keypain | Kids&fen
| Kikilofen | Lamidon | Lapin kids | Lc phen | Lidopain | Logphen | Lupien | Mcfen | Medicofen |
Mediphen | Medrapen | Migafen fresh soft | Migfen | Monitin | New ibufen | Onopain | Pain angel
ibu | Panaphen | Paren | Penbon | Penzal w ibuprofen | Phil ibuprofen | Picopen | Pofen | Polyene
| Prufen | Q sen | Rheumedon | Rheutis | Rifen | Riorupen | Rofen | Rupien | Rurupino | S vil ibu
| Scopen | Speedic | Speedpain nano | Speedpen nano | Tacenol ibu | Taksen 400 ibuprofen |
Tatanal | Trisfen | Ttotto | Upotan | Wannabe olive | Yuhan ibuprofen | Zeropain | Zorfen;
- (KW) Kuwait: Advil | Balkaprofen | Brufen | Fenbid | Fleximex | I profen | Ibucare | Ibugesic | Irfen
| Jazofen | Kendo | Omafen | Prof | Profinal | Sapofen | Saridon N;
 - (LB) Lebanon: Algifast pediatric | Apo ibuprofen | Brufen | Captain | Dimidon | Doloraz | Febridex
| Fleximex | Ibugesic | Ibusol | Pedafen | Pedifen | Perofen | Profinal | Sapofen;
 - (LT) Lithuania: Brumare | Deliprom | Ibufanti | Ibufen | Ibufug | Ibugard | Ibumetin | Ibuprom |
Ibuprom express | Ibusan | Motrin | Nurofen | Nurofen forte | Nypren;
 - (LU) Luxembourg: Brufen | Ibu | Ibuhexal | Ibuprofen ct | Ibuprofen eg | Ibuprofen sandoz | Imbun
| Malafene | Nurofen | Optalidon | Schmerz dolgit | Solufen;
 - (LV) Latvia: Apo ibuprofen | Brufen | Brumare | Burana | Dolormin migraene | Forsium | Ibufen |
Ibugard | Ibugesic | Ibumax | Ibuprof | Ibusan | Ibustar kids | Irfen | Motrin | Neoprofen | Nurofen
| Nurofen berniem | Nypren;
 - (MA) Morocco: Agifene | Algantil | Algotene | Brufen | Dolifen | Ibumac | Intralgis | No dol fen |
Nurodol | Prodol | Trifene;
 - (MX) Mexico: Actron | Adopren | Aavil | Aavil 12 horas | Advil infantil | Aavil max | Afleno | Aflusil
| Ainex | Aldofen | Algidol | Arbufentab | Bestafen | Brixsons | Butacortelone | Cadoil | Carone |
Citalgan | Days | Defixod | Dipofen | Diprodol | Dolprin | Dolprofen | Dolver | Dulfan | Eufenil |
Eufenil forte | Febratic | Fidoin q | Gelidol | Gelubrin | Gobrosan | Ibuflam | Ibuflex | Ibuprofeno |
Ifentil | Infantil motrin | Inpained | Jarisoft | Medifen | Mejorultra | Motiflex | Motrin | Nafendol |
Neo melufen | Neobrufen | Nuropren | Piradol | Pro xb | Probuxil | Quadrax | Realdrax | Sindolen
| Tabalon | Temptra fen | Viczen | Zoafalet h;
 - (MY) Malaysia: Axcel ibuprofen | Brufen | Brusic | Cp pharma ibuprofen | Evofen | Ibufac | Ibufen
| Inflafen | Nurofen express | Profinal | Prufen | Winofen | Zorafen;
 - (NG) Nigeria: Alapharm ibuprofen | Alben ibuprofen | Astropen | Avrofen ibuprofen | Babuflam
| Benprofen | Bio profen | Bioraj ibuprofen | Bkb vibumol | Bond ibuprofen | Brufen | Brustan n
| Bucaldol | Bulprofen | Chemofen | Children's ibuprofen | Dana lam | Danbufen | Darprofen |
Drugzone ibuprofen | Elbufen | Eminent drugs ibuprofen | Emprofen | Epl ibuprofen | Esefen |
Esen | Eurofen | Fatibufen | Friendship ibuprofen | Galen's ibuprofen | Garyfen | Gauze ibuprofen
| Genafen | Gilgal ibuprofen | Glopafen | Grapofen | Ibuage | Ibuflam | Ibumed | Ibupain | Ibupkris
| Ibuprocel | Ibuprofen bp | Ibusar | Ibuten | Ibuvil | Jesifen | Jopan profen | Jsp ibuprofen | Kayhelt
ibuprofen | Leyfen | M & b ibuprofen | Makupar | Multifen | Nalfen forte | Napfen | New divine
vibumol | Norprofen | Nosak ibuprofen | Nurofen advance | Nurofen express | Obnoprofen |
Ocprofen | Pal ibuprofen | Peace ibuprofen | Piccan | Shreefen | Softhealth ibuprofen | Stevufen
| Sunymet ibuprofen | Tabalon | Tabufen | Talibex ibuprofen | Taskine | Tramibu | Ultfen | Vemfen
| Vpl ibuprofen | Westmed ibuprofen | Zedfen kid | Zestefen | Zonason ibuprofen;
 - (NL) Netherlands: Aavil reliva | Albert heijn ibuprofen | Antiruggine | Brufen | Ibosure | Ibuprofen
actavis | Ibuprofen apotex | Ibuprofen htp | Ibuprofen linn | Ibuprofen sandoz | Ibuprofen teva
| Jumbo ibuprofen | Kruidvat ibuprofen | Leidapharm ibuprofen | Nurofen | Nurofen fastine |
Nurofen fastine zavance;
 - (NO) Norway: Brufen | Ibugard | Ibumetin | Ibuprox | Ibox | Neoprofen | Nurofen | Nurofen forte
| Nurofen junior | Nurofenpro | Pedea;
 - (NZ) New Zealand: Advil | Brufen | Ethics ibuprofen | Fenapaed | Ibuprof | Ibuprofen apotex |
Nurofen | Nurofen for children | Nurofen for children 5-12 years | Nurofen zavance | Pharmacy
health ibuprofen | Your pharmacy ibuprofen;
 - (OM) Oman: Omafen;
 - (PA) Panama: Dorival | Febrin | Ibumer | Ibupharma | Ibuprofeno | Ibuprofeno mk | Ipson | Ipson
forte | Multidol | Pedea | Regulador gesteira;
 - (PE) Peru: Afebril | Alfadol | Alivium | Anafidol | Anaflex mujer NF | Apo ibuprofen | Banes | Banes
forte | Biodol | Bren | Doladol | Dolfarb | Dolin forte | Dolobay kids | Dolochek plus | Dolodif kids
| Doloflam extraforte | Dolokid s | Dolomax | Dolomil | Dolonet | Doloral | Doloral forte | Dorelin |

- Ebufac | Febryn | Flamadol | Flamalgésico | Fredol | Gesic | Gofen | Ibufas | Ibufenol | Ibumejoral | Ibupain | Ibupirac | Ibupiról | Ibuprofeno | Ibupromil c | Ibusol | Ibuzone | Lefebren | Macrogesic | Mecodolfen | Mecopiron | Mediflam | Mediflam niños | Metradol | Midol rapid action | Molargésico | Motrin | Multigesico | Natudol | Nayflex mujer | Nonpiron | Pedeá | Pironal | Pirovan | Plidocheck plus | Profen kids | Profinal | Provon | Saldeva | Terbofen;
- (PH) Philippines: Advil | Anvil | Brufen | Bufen | Dolafen | Dolan fp | Fasgesic | FeverFree | Fevral | Ibuped | Idyl | Medbufen | Medicol i | Morexen | Mutrim | Myvil | Nurofen | Penfro | Profen | Rheuxan | Saphfen | Skelan | Sqfen | Ynex;
 - (PK) Pakistan: Aksofen | Algofen | Ambrofen | Amofen | Anglofen | Azofin | B fen DS | Baprofen | Baxfen | Befen | Bellfen | Blutol | Blutol ds | Blufen | Boschofen | Bril | Bril ds | Brofanic | Brofanic ds | Brucam | Brufedin | Brufen | Bruflet | Brugesic | Brunac | Brusan | Bruwell | Budex ds | Deltafen | Dolofen | Dorafen | Eprofen | Evagesic | Fenbro | Fufen | Glare ds | Glitfen | Hugon | Ibtol | Ibu profen ds | Ibu slow | Ibucil | Ibufen | Ibufin | Ibugesic | Ibulis | Ibumid | Ibumor | Ibuperk | Ibuquik | Iburin | Ibuser | Ibusim ds | Ifen | Inflam | Inzel | Iodofen | Jaefen | Jazofen | Karbufen | Kata ds | Legesic | Mayfanax | Meblot | Meblot sr | Medfen | Mefen | Megafen | Mufen | Mybrofen | Newibu | Newibu ds | Novofen | Nuprin | Nuprofen | Nurofen ds | Ocefén | Ofen | Ottofen | Pabrofen | Pango | Pedifen | Pharfen | Profen | Promin | Prufie | Pyrofen | Radifen | Ramafen | Rapifen | Rebugen | Regofen | Resfen | Rhunor | Ribunal | Rofen | Ruberin | Rumafen | Rumin | Selprofen | Semorfen | Skifen | Synofen | Tabrofen | Tagafin | Torfen | True-fen | Werridal | Wilfen | Winkidofen ds | Xaleve | Xiben | Zayfen | Zefen | Zofen | Zufen;
 - (PL) Poland: Axoprofen | Axoprofen forte | Babyfen | Bolinet | Brufen | Bufenik | Ibalgin | Ibufen | Ibufen d | Ibum | Ibum comfort | Ibum express | Ibum express forte | Ibum femina | Ibum forte | Ibum forte minicaps | Ibum sprint | Ibum Supermax | Ibuprofen banner | Ibuprofen Dr. Max | Ibuprofen hasco | Ibuprofen max aurovitas | Ibuprom dla dzieci | Ibuprom max sprint | Ibutact | Ibuxal | Ifenin dla dzieci | Kidofen | Kidofen max | Metafen ibuprofen | Mig | Milifen | Nurofen | Nurofen dla dzieci Forte | Nurofen express femina | Nurofen express forte | Nurofen menstrual | Paduden express | Pediprofen;
 - (PR) Puerto Rico: Aavil | Children's ibuprofen | Infants Ibuprofen | Menadol | Motrin | Motrin childs | Motrin ib | Motrin infants | Proprinal;
 - (PT) Portugal: Arfen | Brufen | Brufen liq | Danofen | Dimidon | Dolocyl | Ib u ron | Ibalgin | Ibudol | Ibuprofeno | Ibuprofeno algik | Ibuprofeno farmoz | Ibuprofeno generis | Ibuprofeno Pharmakern | Imofen | Inabrin | Inoflon | Moment | Motrin | Nurofen xpress | Nurofen zavance | Pedeá | Solufen | Stodorfen | Trifene | Zip a dol;
 - (PY) Paraguay: Actron | Actron pediátrico | Afebril | Afebril forte | Algidol kids | Antidol pediátrico | Banes | Buenox | Calmadol | Carbofeno | Cariodent | Cenflam | Dolgex | Dolgex forte | Dutrifén | Etidol | Fabogesic | Fabogesic niños | Febratic | Finidol | Finidol forte | Finidol plus | Ibu algin | Ibu algin forte | Ibu clerinax | Ibu contradol | Ibu denene | Ibu evanol | Ibu evanol forte | Ibu kids | Ibu metaljina | Ibu temprilan | Ibucler | Ibudolanet | Ibufil | Ibuflam | Ibuflam pediátrico | Ibugesic | Ibulib | Ibumetaljina | Ibunest | Ibupan | Ibupirac | Ibupirac a.p. | Ibuprofeno caplin point | Ibuprofeno catedral | Ibuprofeno cellofarm | Ibuprofeno eximfarma | Ibuprofeno genfar | Ibuprofeno medinac | Ibuprofeno millet | Ibuprofeno p.a.n | Ibuprofeno vivele | Ibusol | Ibusol forte | Ibusol pediátrico | Ibuterm | Ipson | Ipson forte | Kevadon ibuprofeno | Kitadol pediátrico | Kitadol rapid | Movidol ibu | Niofen | Niofen forte | Niofen lc | Pediaprofen | Pediaprofen forte | Profenol | Sindol | Somadol | Vantril | Vantril retard;
 - (QA) Qatar: Advil | Aavil Liqui-Gels | Apo-Ibuprofen | Brufen | Brufen Retard | Brus | Caldolor | Corelar | Dolgit | Doloraz | Dolormin Extra | Fen-Q | I-Profen | Ibu | Ibufen | Ibugesic | Nureflex | Nurofen Children | Omafen | Panafen | Pedeá | Pediaver | Perofen | Prof | Profinal | Profinal Paediatric | Remofen | Riaprofen;
 - (RO) Romania: Abfen | Adagin | Advil ultra forte | Algin baby | Arbifen | Biofen | Biofen forte | Brufen | Brufen retard | Calmolín arena | Ibalgin baby | Ibalgin express | Ibalgin junior | Ibuprofen arena | Ibuprofen dr.max | Ibuprofen Heaton | Ibuprofen nutra | Ibuprom | Ibuvalen | Inflanor | Inflanor forte | Marcofen | Nurofen express | Nurofen immedia ultra | Nurofen junior | Paduden | Paduden express | Paduden express forte | Panafen;
 - (RU) Russian Federation: Advil | Aavil for children | Advil maximum | Apo ibuprofen | Bolinet | Brufen | Bumidol | Ibufen | Ibuprofen extracap | Ibuprofen hemofarm | Ibuprom sprint | Maxicold for child | Mig for babies | Mig for children | Nebolin | Next uno express | Next uno express forte

- | Novigan | Nurofactor | Nurofen | Nurofen express | Nurofen express forte | Nurofen for children | Nurofen ultracap | Nurofen ultracap forte | Pedea | Sedalgin sprint;
- (SA) Saudi Arabia: Brufen | Children's aadvil | Emifen | Fenbid | Ibugesic | Jazofen | Junifen | Medafen | Nufen | Nurofen | Prof | Profinal | Rapifen | Sapofen;
- (SE) Sweden: Brufen | Burana | Ibuprofen | Ibuprofen abece | Ibuprofen apofri | Ipren | Ipressa | Nurofen | Nurofen jordgubb;
- (SG) Singapore: Bifen | Brufen | Ibuloide | Nurofen | Nurofen express | Sugafen;
- (SI) Slovenia: Brufen | Ibubel | Ibuprofen inn farm;
- (SK) Slovakia: Aadvil | Apo ibuprofen | Brufen | Dolgit | Ibalgin | Ibalgin rapidcaps | Ibuprofen dr.max | Ibuprofen farmalider | Ibuprofen rapid apotex | Ibuprofen stada | Nurofen | Nurofen pre deti | Panactiv | Pedea | Solpaflex | Tempasa;
- (SL) Sierra Leone: Actiflam | Ibuprofene;
- (SR) Suriname: Advil childrens | Brufen | Bufedon | Children's motrin | Da ibuprofen | Gobrosan | Ibugesic | Ibugesic sr | Ibuprofen apotex | Nurofen | Profen;
- (SV) El Salvador: Ibufeno;
- (TH) Thailand: Acprofen | Ambufen | Anbifen | Aprofen | Babefen | Bethofen | Borafen | Borakid | Brofen | Brufen | Brugin | Brumed | Bruprin | Brusil | Brusoft | Buflex | Bumed | Bunofen | Buprofen | Burafen | Busofen | Cefen | Coprofen paediatric | Deflem | Duran | Fafen | Fob fen | Forbafen | G-fen | Greatofen | Heidi | I Fen | I profen | Ibo | Ibrofen | Ibru | Ibucin | Ibucin R | Ibufac | Ibufex Paediatric | Ibugesic | Ibukids | Ibulan | Ibuman | Iburen | Junimol | Kidifen | Mano Bruzone | Nurofen zavance | Omnifen | Ostofen | Penofen | Pharmafen | Pippen | Prefen | Profen | Profena | Profeno | Profensic | Rufan | Ruprofen | Schufen | Suphen | Tofen | Tonifen | Trofen | Umafen | Venofen;
- (TN) Tunisia: Adofen | Balkaprofen | Balkaprofene | Brufen | Buprofen | Cycloff | Dolven | Gelufene | Ibuphil | Moment | Novafen | Pedea;
- (TR) Turkey: Apireks | Brufen | Bruprex | Brus | Cicifen | Ibu | Ibu-fort | Ibufen | Ibutrit | Pedifen | Profen | Profen fort | Repoza | Rofen | Siyafen | Ultrafen | Upren;
- (TW) Taiwan: Anflagen | Arfen | Artofen | Brofen | Buburone | Bufen | Buprofen | Easifon | Guasheaton | Ibu | Ibufen | Ibulin | Idofen | Illume | Iprofen | Iprofen | Iputon | Kortufen | Librofem | Mobufen | Neoprofen | Nobafon | Nobgen | Outinflamm | Profen | Profenil | Sconin | Scoton | Serviprofen | Shouenlin | Suyoubao ib | Tarein | Tolax | Yonfupharm ibuprofen;
- (UA) Ukraine: Affida express | Affida max express | Blokmax for kids | Bofen | Bolinet | Brufen | Brufen rapid | Burana | Darfen express | Darfen kids | Darfen kids forte | Darfen ultracap | Eurofast | Eurofast express | Gofen | Ibufen | Ibufen d | Ibufen for children | Ibufen forte | Ibufen junior | Ibunorm | Ibunorm baby | Ibuprofen baby | Ibuprofen darnitsa | Ibuprom extra | Ibuprom sprint | Ibuprom sprint max | Ibupron | Ibutard | Imet | Imet 2 | Imet 4 | Novigan | Novo profen | Nurofen | Nurofen express | Nurofen express forte | Nurofen for children | Nurofen for children forte | Nurofen kids forte | Orafen | Solpaflex | Termidol | Teufen;
- (UG) Uganda: Agofen | Asfen fc | Axcel ibuprofen | Betafen | Brumed | Fenpaed | Flamo | Gofen | Ibucach | Ibumex | Iburen | K fen | Medifen | Nurofen | Nurofen express | Nurofen for children | Simfen;
- (UY) Uruguay: Actron | Actron pediatrico | Argifen | Bioprofen | Fabogesic | Fabogesic ninos | Febratic | Fluxus | Ibubenitol | Ibuevanol | Ibuevanol rapida accion | Ibumultin | Ibupirac | Ibupirac fast | Ibupirac pediatrico | Ibupiretas | Ibuprofeno | Ibuprofeno szabo | Ibusol | Ibutenk | Perifar | Perifar infantil | Ponstinetas | Privalgia | Prontopiren | Zolpirol NF;
- (VE) Venezuela, Bolivarian Republic of: Anapir | Bioprofen | Brane | Brugesic | Buprodol | Disbafen maxx | Dologesic | Dologesic fem | Femcaps | Femmexplus | Ibucaps | Ibufen | Ibumax | Ibun | Ibuprin | Ibuprofeno | Ibuprofeno-gc | Ibuprofeno-penta | Ibuvin | Ibuwin forte | Iporet | Iprofene | Lumbax | Maydol | Mestral | Metradol | Motrin | Nanprofen | Pedibu;
- (VN) Viet Nam: A.t ibuprofen | Agirofen | Bufecol | Iratac | Polebafen | Sosfever fort;
- (YE) Yemen: Ibofen | Ibofen s.r.;
- (ZA) South Africa: Adco-Ibuprofen | Brufen | Ibucine | Ibugesic | Ibumed | Inflamax paediatric | Nurofen | Pedea | Rolab-ibuprofen;
- (ZM) Zambia: Actiflam | Bren | Butafen | Ibu | Ibufen | Ibufil | Ibufamar | Ibukant | Ibumex | Ibunate | Intafen | Norswel | Novafen | Nurofen | Nurofen for children | Nurofen for children strawberry | Pedifen | Profen | Profundin | Rufen | Rumifen | Sirofen;

- (ZW) Zimbabwe: Dolex | Nurofen | Nurofen express | Nurofen for children

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